[BellaBot Pro]Deployment Manual

Revision Records

Revised by	Time of revision	Revised content	Current version number
Lele CHANG	2026.07.02	Upgrade the format and contents	V 1.0

Introduction
This manual provides standardized deployment guidelines for Pudu Robotics BellaBot Pro robot, targeting project implementation, on-site operation and maintenance, and equipment management personnel, ensuring the robot is deployed compliantly and efficiently, and operates stably online.

1.Requirement communication

To improve service quality and customer experience, it is recommended to communicate with the customer in advance and confirm the following:
● Clarify initial customer requirements and simultaneously obtain information such as the robot's operating area, charging pile location, and the number of elevators the robot needs to use;
● The system collects on-site environmental parameters (including ground material, obstacle distribution, dynamic changes in the scene, etc.), and preliminarily assesses the precautions for on-site deployment;
● Based on customer requirements and environmental assessment results, analyze potential environmental adaptability modification requirements and formulate a preliminary implementation plan.

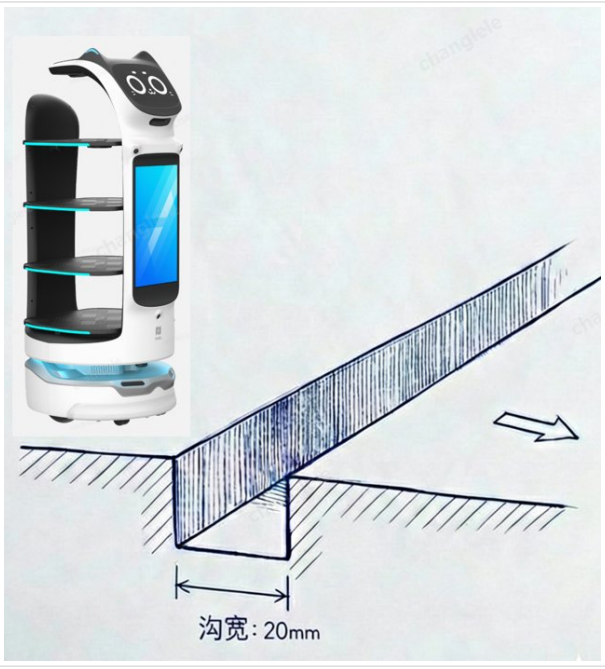
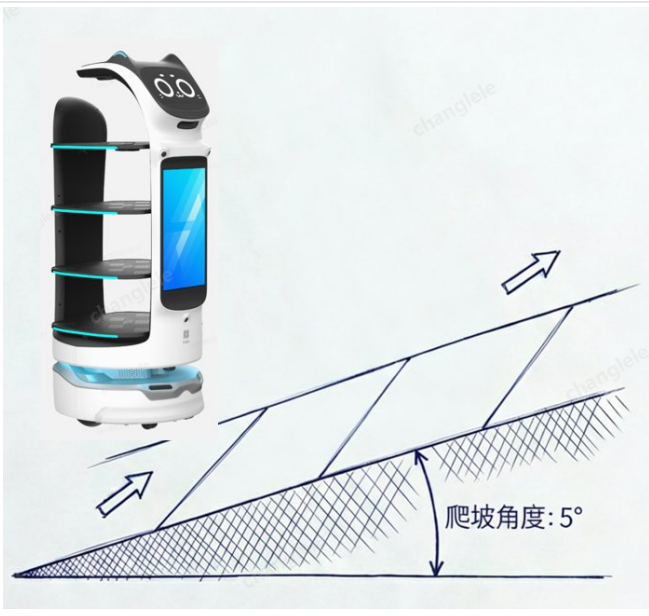
2.Environment assessment

2.1 Vertical space assessment

Minimum pass-through width: 65 cm

2.2 Ground assessment

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	Crossing height	Gap width	climbing slope
Illustration			
Maximum pass-through value	5 mm	20 mm	5°

3.Deployment Preparations

3.1 Unpacking and inspection

3.1.1 Outer packaging inspection

Before unboxing, please check that the product's outer packaging is intact. If there is any damage, please record it with a video and contact a PUDU technical engineer promptly.



3.1.2 Appearance check of robot and components

- Check the appearance of the robot and its sensors to ensure that the structure is intact and undamaged.
- If there is any damage, please record it with a video and contact a PUDU technical engineer promptly;
- If there is only dirt on the surface, please wipe it clean with a dust-free cloth.

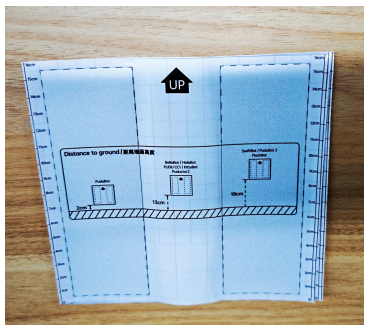
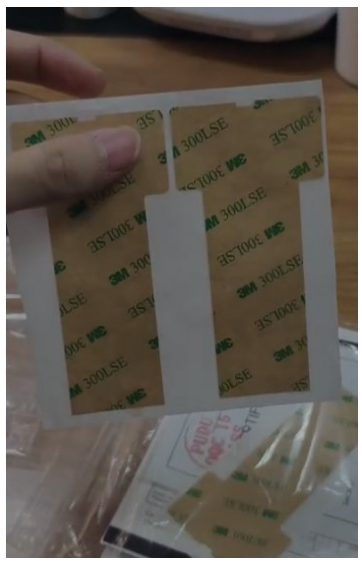
The locations of the robot's sensors and main components are shown in the figure below:



3.1.3 Material inspection

After unpacking the robot, inspect the materials.

Serial number	Name	Illustration	Quantity
1	Complete unit	/	1
2	Charger	/	1

3	Start/stop positioning sticker		1
4	User manual	/	1
5	Anti-drop reflector		1 (package)
6	Tray adjustment port cover adhesive		1 (package)

3.1.4 Remove the protective film

- Remove the protective film from the robot's screen to ensure touch screen sensitivity.
- Remove the protective film from the top-view VSLAM camera lens to ensure the fill light function works properly.





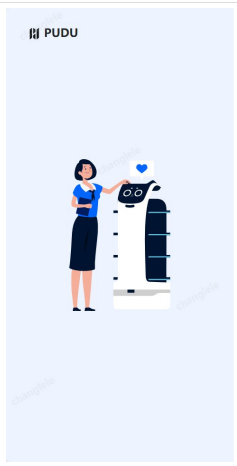
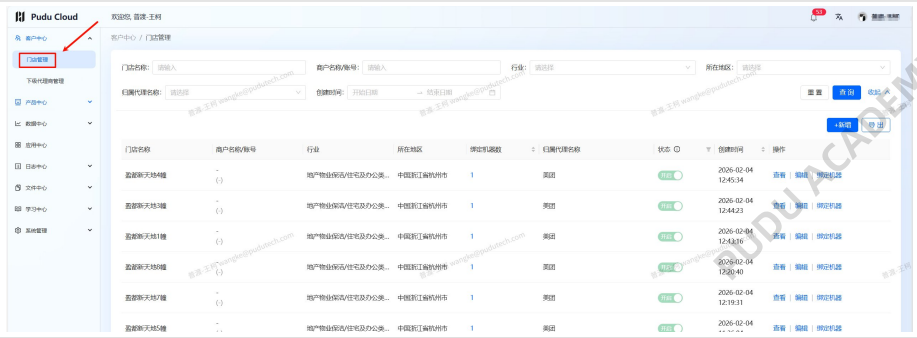
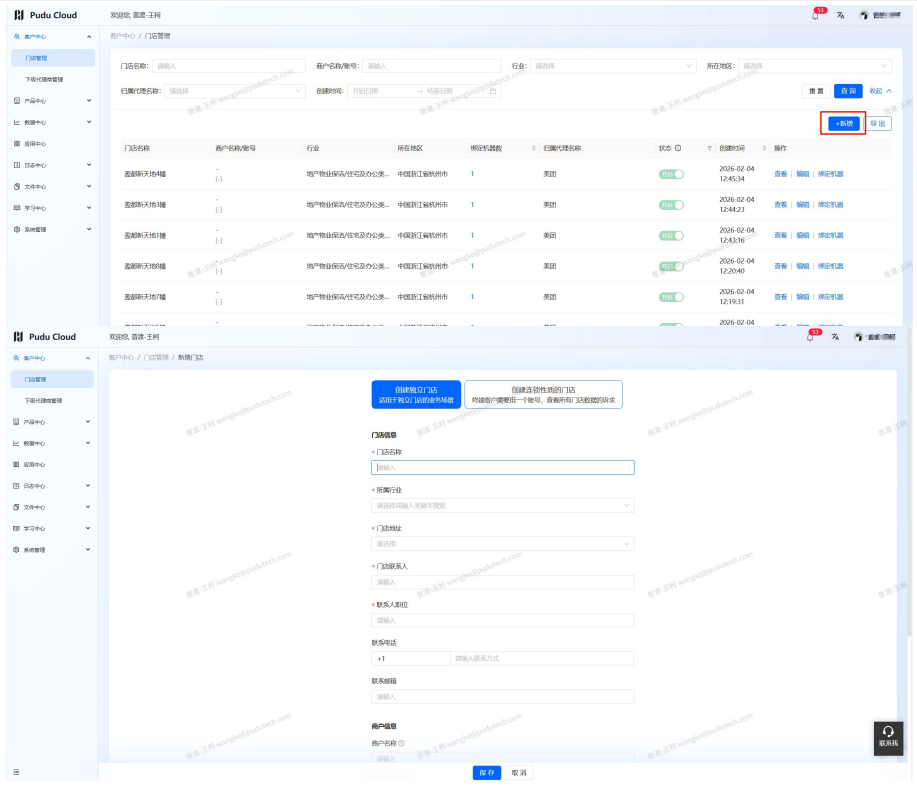
3.2 Power replenishment

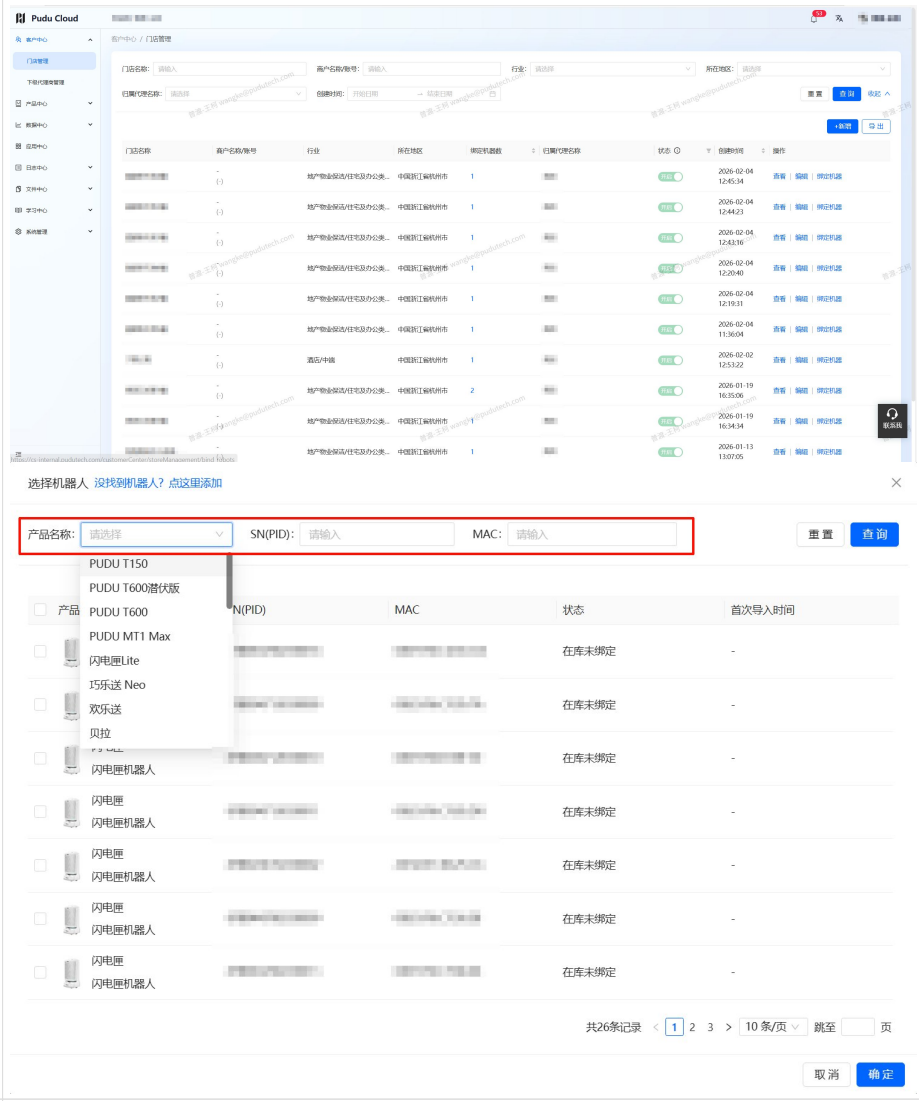
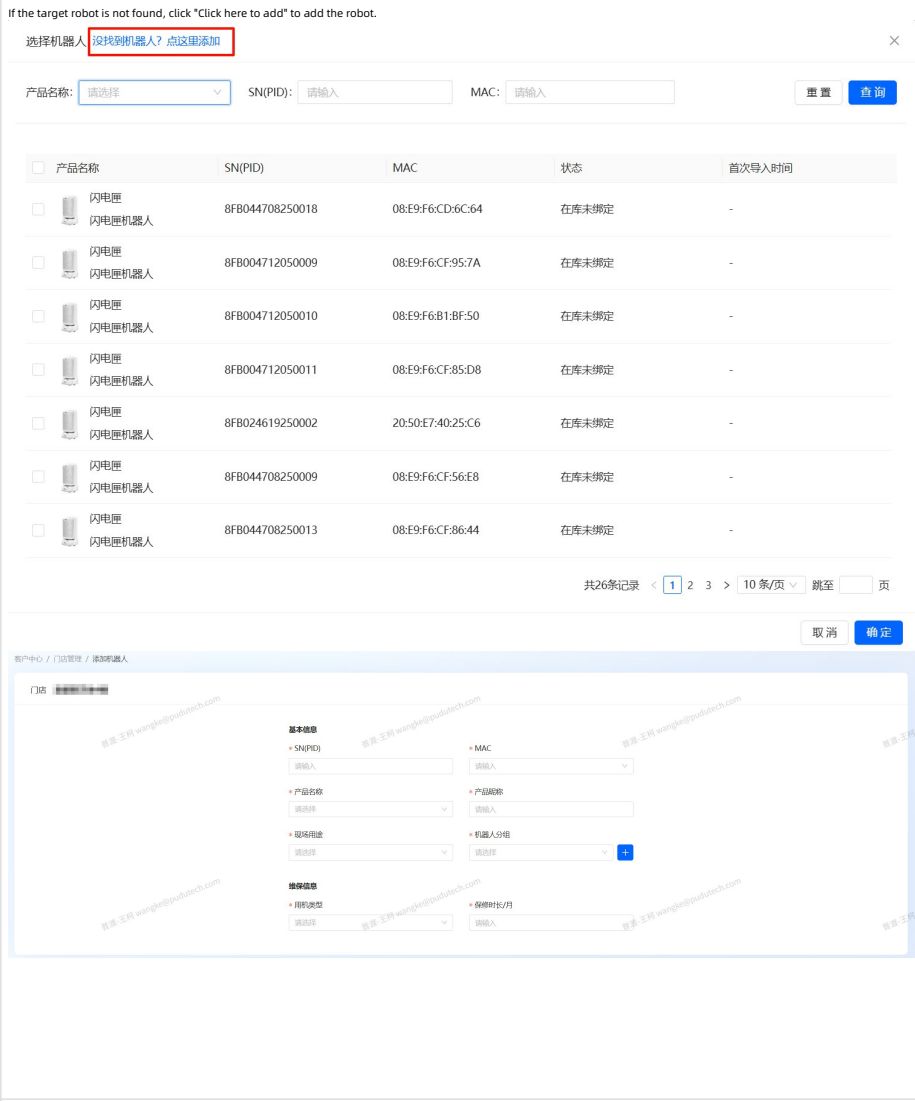
To ensure battery performance, fully charge the battery before using the robot for the first time to activate the battery.

3.3 Activate the robot

The table below uses binding a store to a dealer as an example.

Steps	How to operate	Reference diagram	Remark

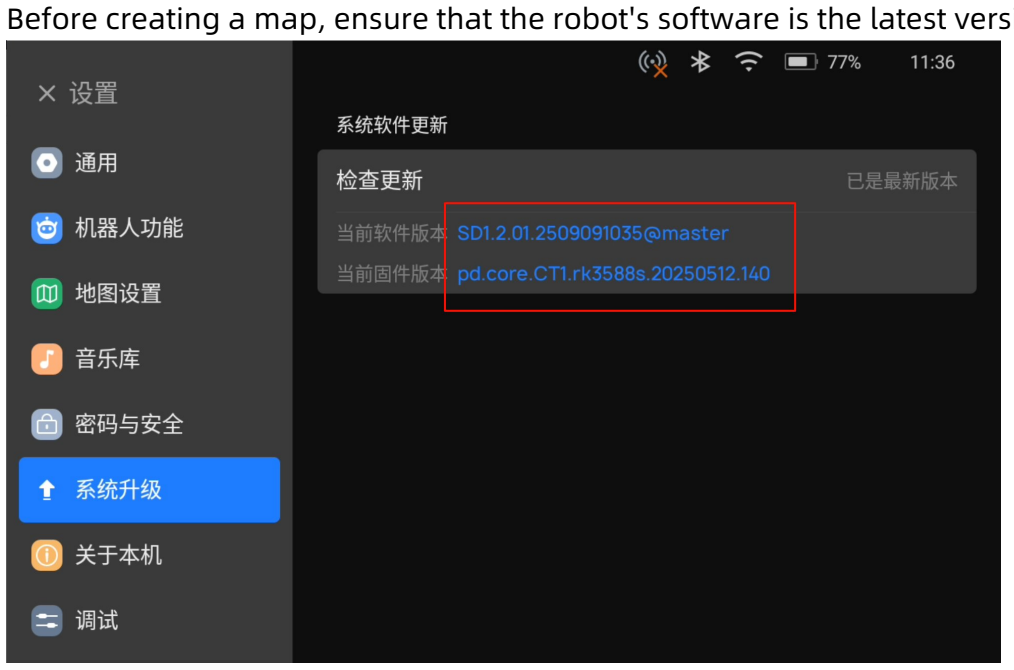
① Turn on the robot	Press and hold the power button for 3 seconds, and let go after the light strip lights up.		After turning on the robot, the robot screen will light up and display a power-on animation.
② Record the MAC address	Follow the on-screen instructions to complete the language and network configuration, and accurately record the robot's MAC address.		/
③ Log in to the dealer management platform	Open the Pudu Robotics Cloud Platform (Agency) in your computer browser and log in.	 	The login URLs for each region are as follows: ● Chinese mainland: https://cs-internal.pudutech.com ● Germany: https://csg.pudutech.com ● Singapore: https://css.pudutech.com ● United States: https://csu.pudutech.com
④ Go to the store management page	After logging in, click Customer Center > Store Management to enter the store management page.		/
⑤ Create a store	Click Create a store and accurately fill in the relevant store information as prompted on the interface.		After successful creation, you can see the created store in the store list.

⑥ Bind the robot	Click "Bind robot" in the corresponding store entry. In the pop-up "Select robot" window, search for the target robot by product name + MAC address, then click "OK" to complete the binding.		If the target robot is not found, click "Click here to add" to add the robot. 
⑦ Check activation status	Return to the robot to check. Select "Activate". If the robot's screen displays "Activated successfully", the robot has been bound and activated.		/

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3.4 Version update

Before creating a map, ensure that the robot's software is the latest version; otherwise, you need to upgrade it.



4.Create a map

4.1 Mapping introduction

BellaBot Pro provides five mapping methods:



- VSLAM + laser mapping is recommended first.
- If the work map area is small (< 1,000 ㎡), you can choose laser mapping for faster and more efficient operation.
- If markers are already deployed in the venue, prioritize the "Marker (ceiling) + laser" mapping method.

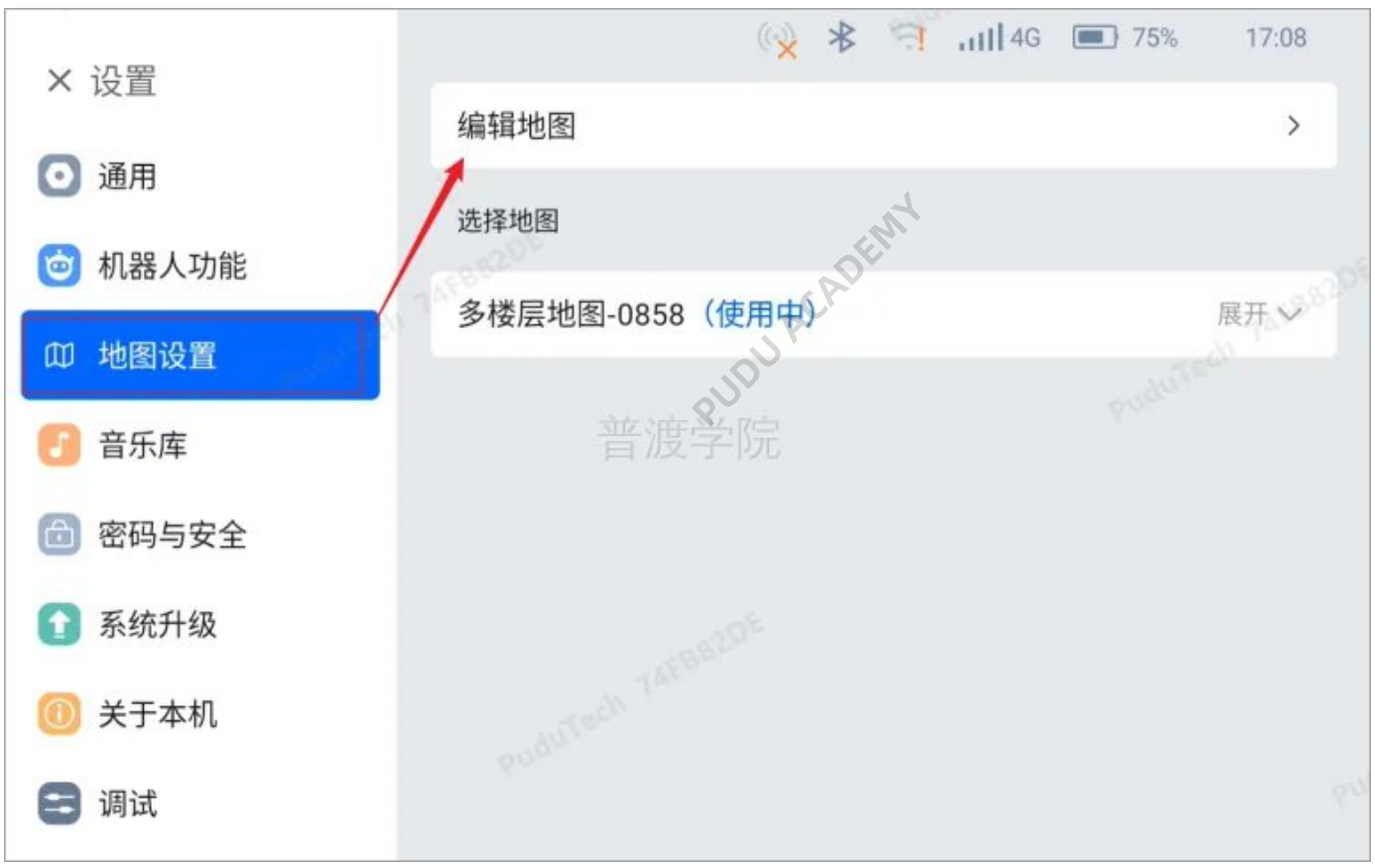
4.2 Mapping steps

To ensure mapping quality, you need to understand the mapping essentials for the following typical scenarios.

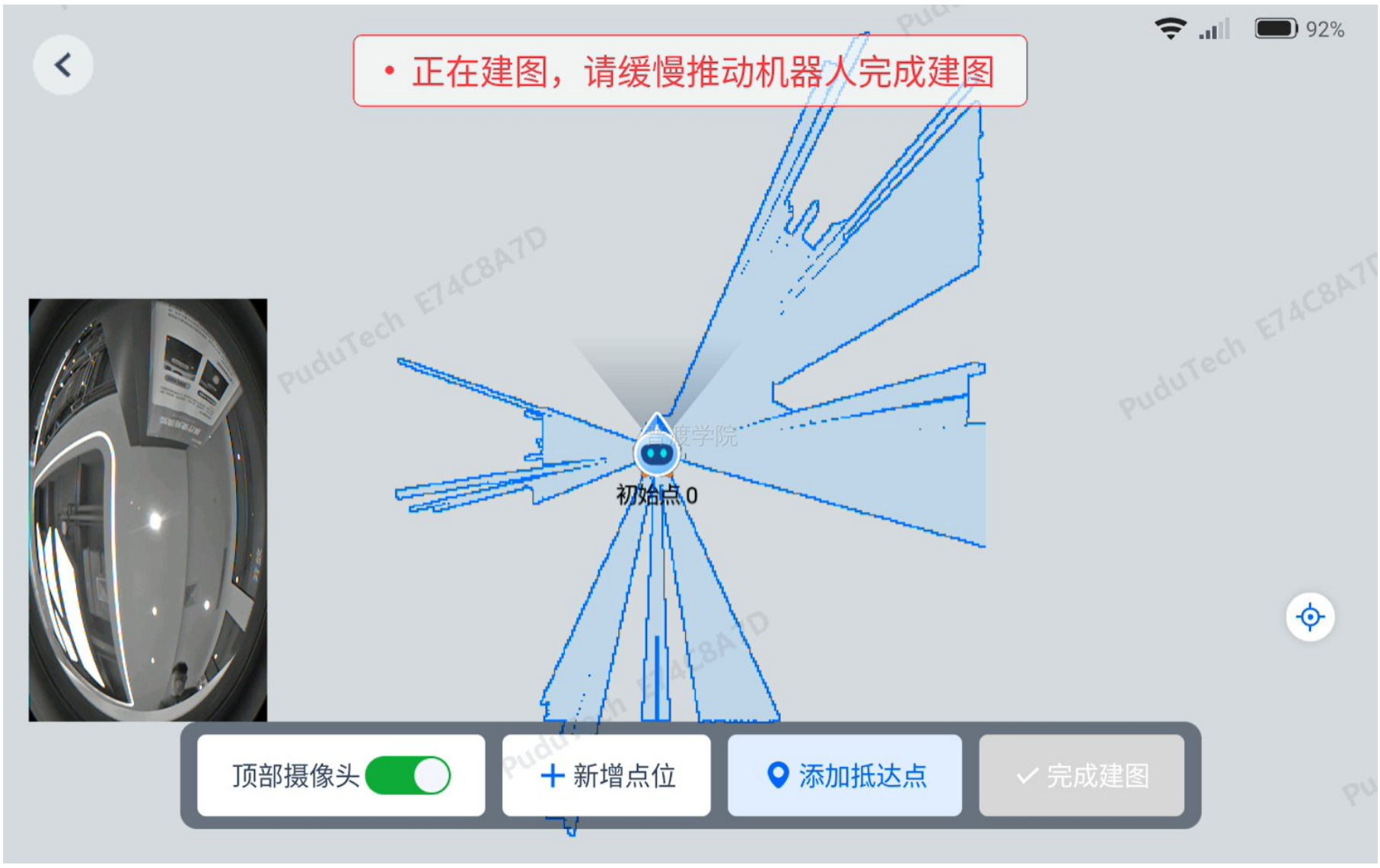
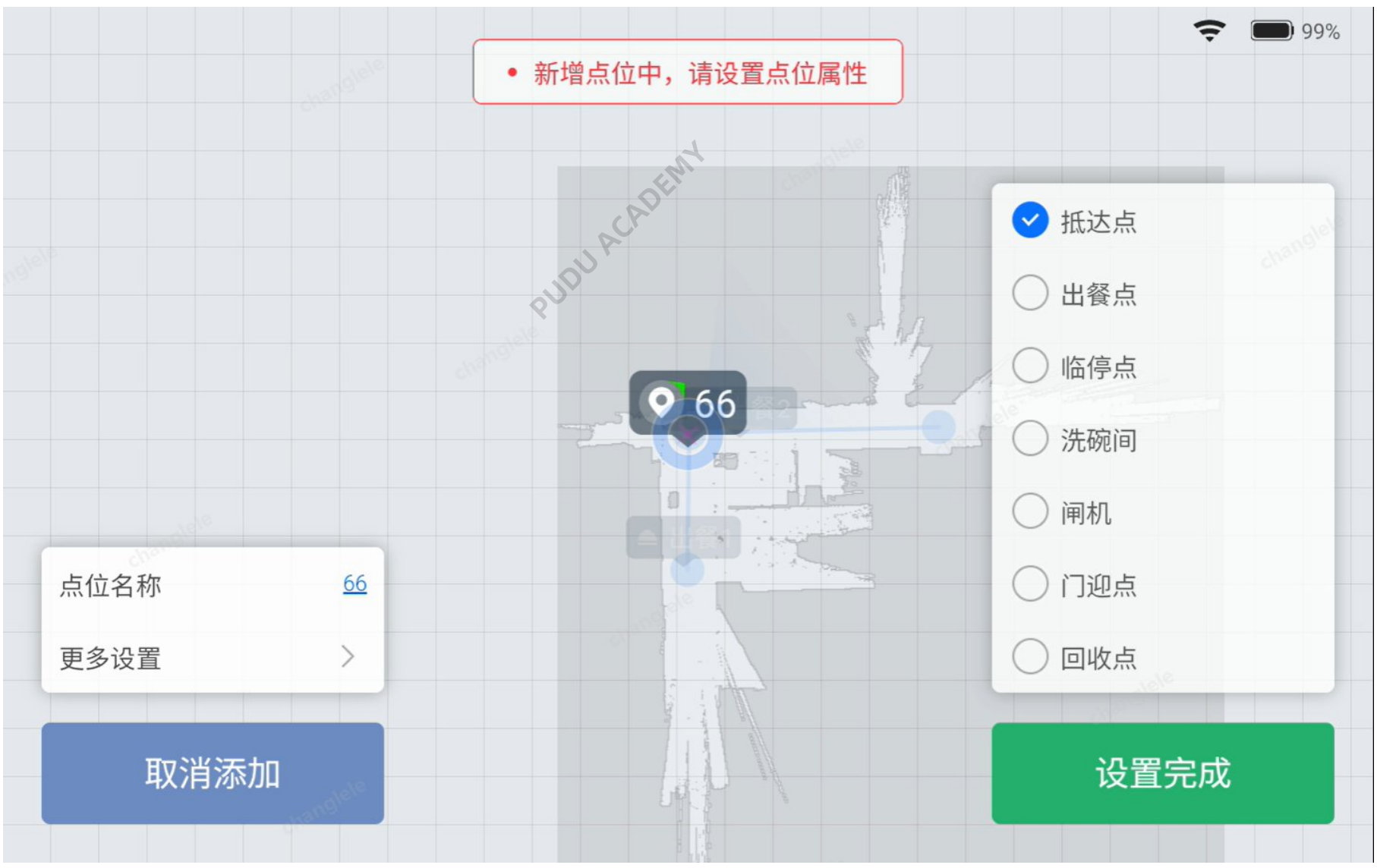
4.2.1 Start mapping

Currently, BellaBot Pro supports creating a single map with a maximum area of 50,000 ㎡.
Taking the "VSLAM + Laser" fusion mapping mode as an example, the specific mapping steps are as follows:

1. When using the robot for the first time, there is no map in the robot. After starting the robot, select "Start Mapping" to open the mapping tool.	

2. If there is already a map in the robot, start the robot and go to the software homepage, then select "Settings."	
3. In the settings menu, select "Map settings," and then select "Edit map" to go to the mapping tool homepage.	

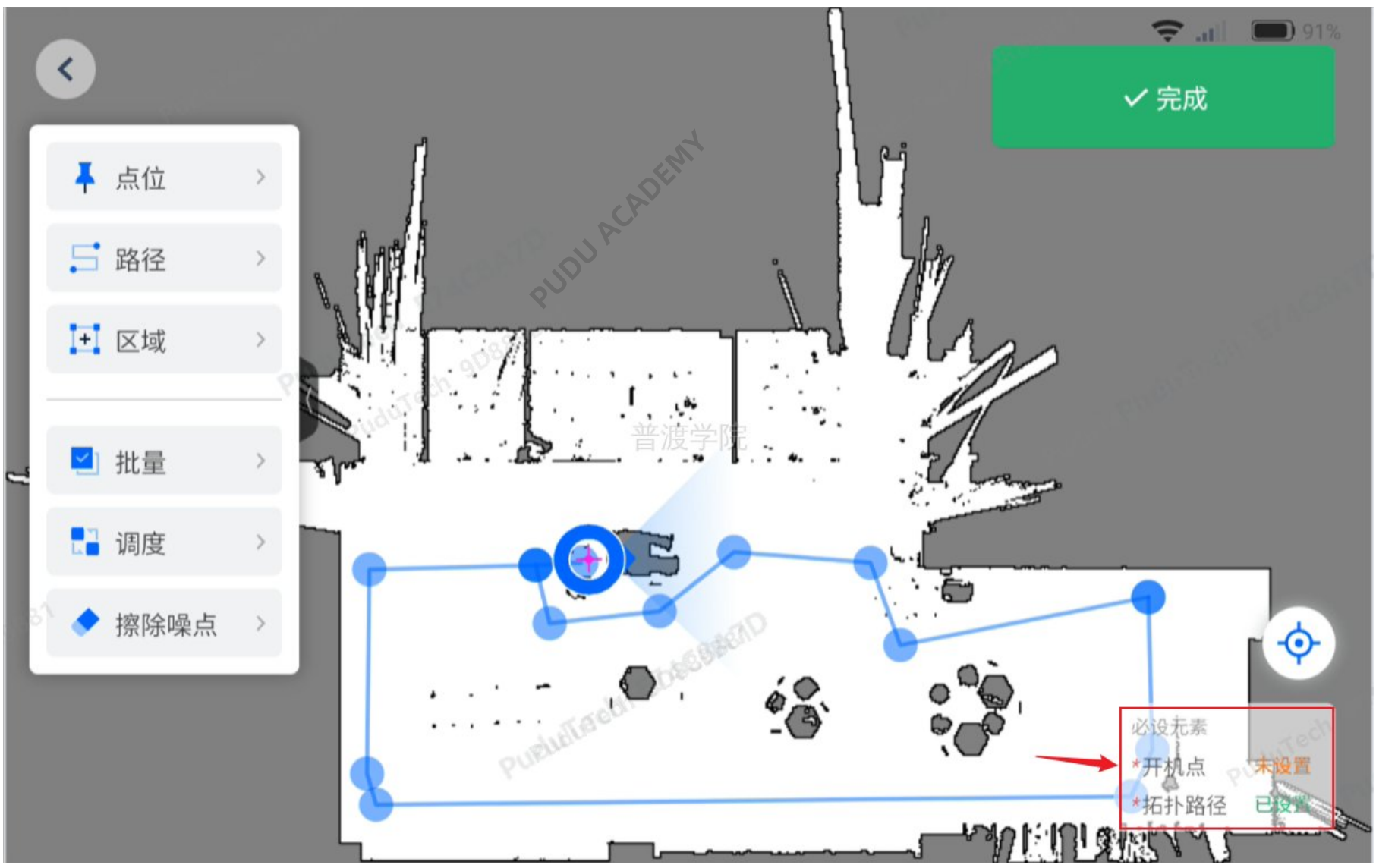
4. On the mapping tool homepage, select "New." Note: On the mapping tool homepage, you can edit, expand, and delete existing maps.	
5. Select the mapping method.	

6. Push the robot to scan the map.	
7. Add corresponding points as required by the scene. After completing the mapping, push the robot back to the starting point. Tip: You can add settings for some points during the mapping process.	

8. Finally, return to the mapping starting point to complete the mapping.



9. After the map is generated, a prompt box will appear in the lower right corner. You need to add a power-on point to save the map.



10. Add a power-on point: Push the robot along the map path to a location that meets the requirements for a power-on point. After selecting "Power-on point," select "Done" and synchronize the map to the cloud platform. This allows multiple robots in the same store to directly reuse the map and prevents local map loss on the robot.



The usage logic for the other four mapping methods is essentially the same as the method described above.

4.2.2 Recommended boot-up points

VSLAM startup point

When using the “VSLAM + laser” mapping method, add VSLAM startup points.

Ceiling with distinct features

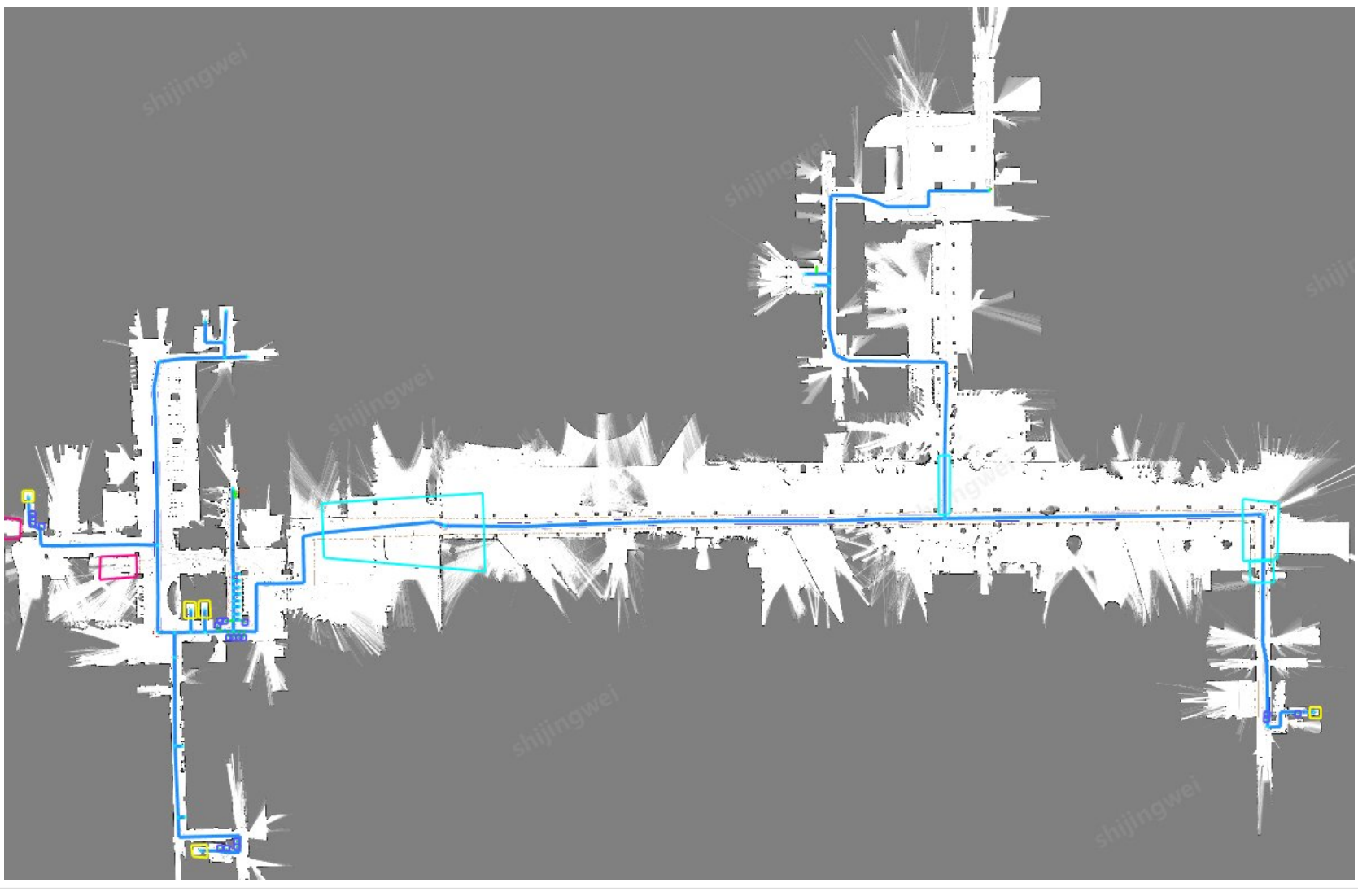


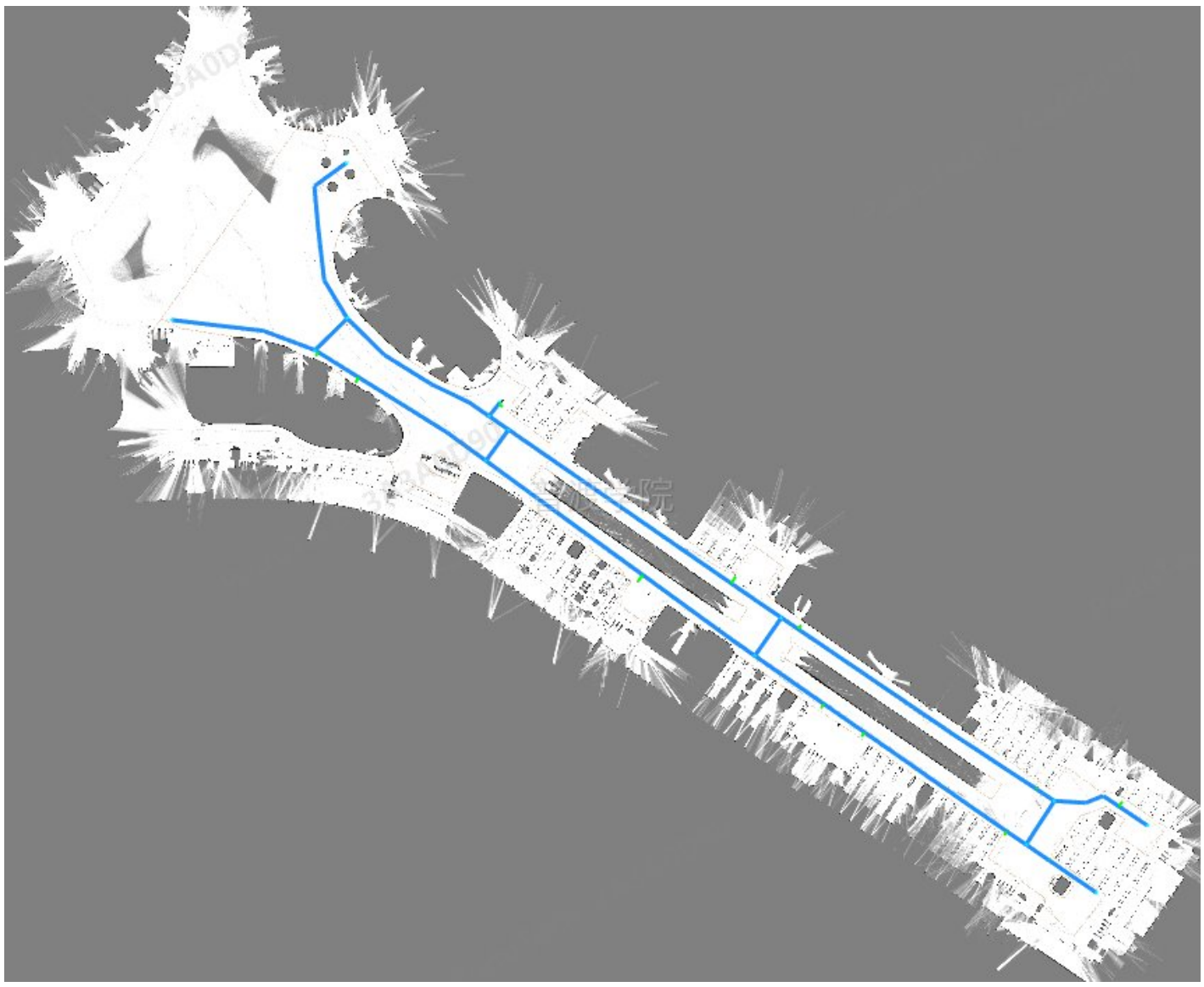
Laser boot-up points

When using "laser mapping", you need to add laser power-on points: the flat environment has distinct features.

A concave surface of a wall corner	A convex surface of a wall corner
	

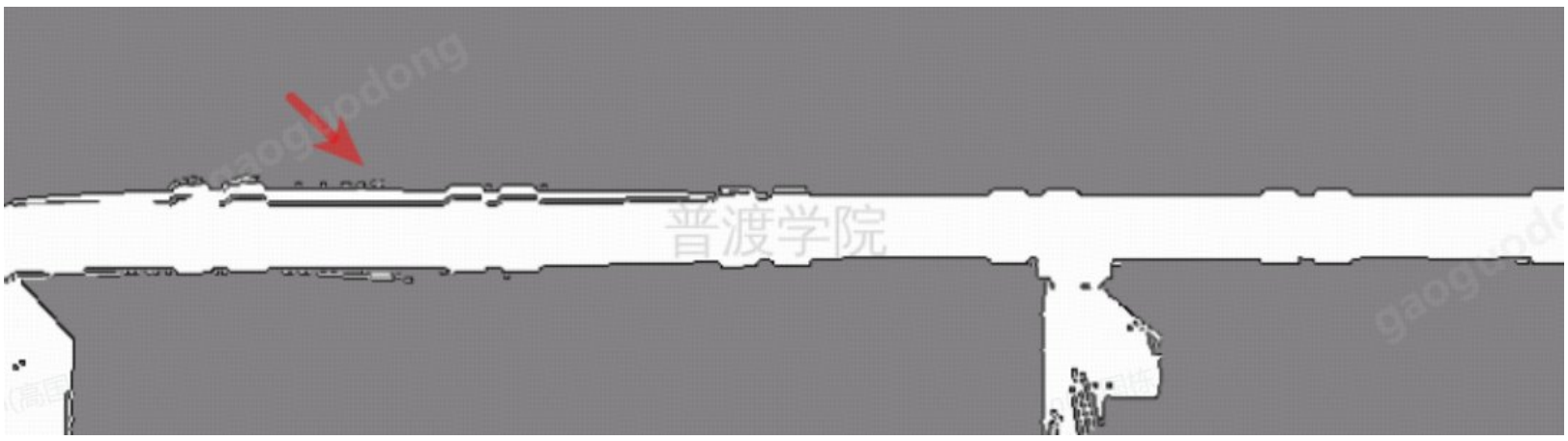
4.2.3 Key points for mapping common scenarios

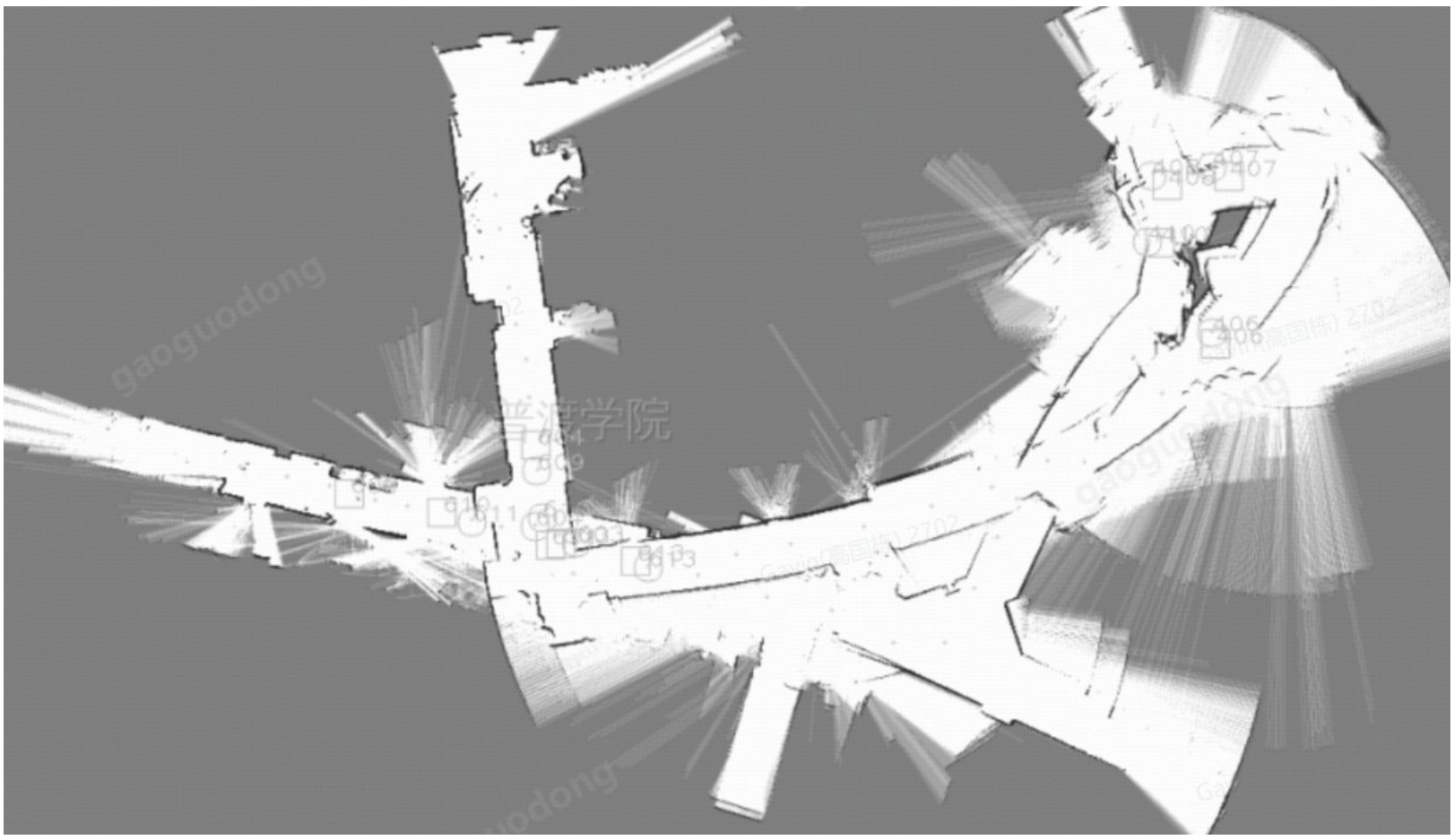
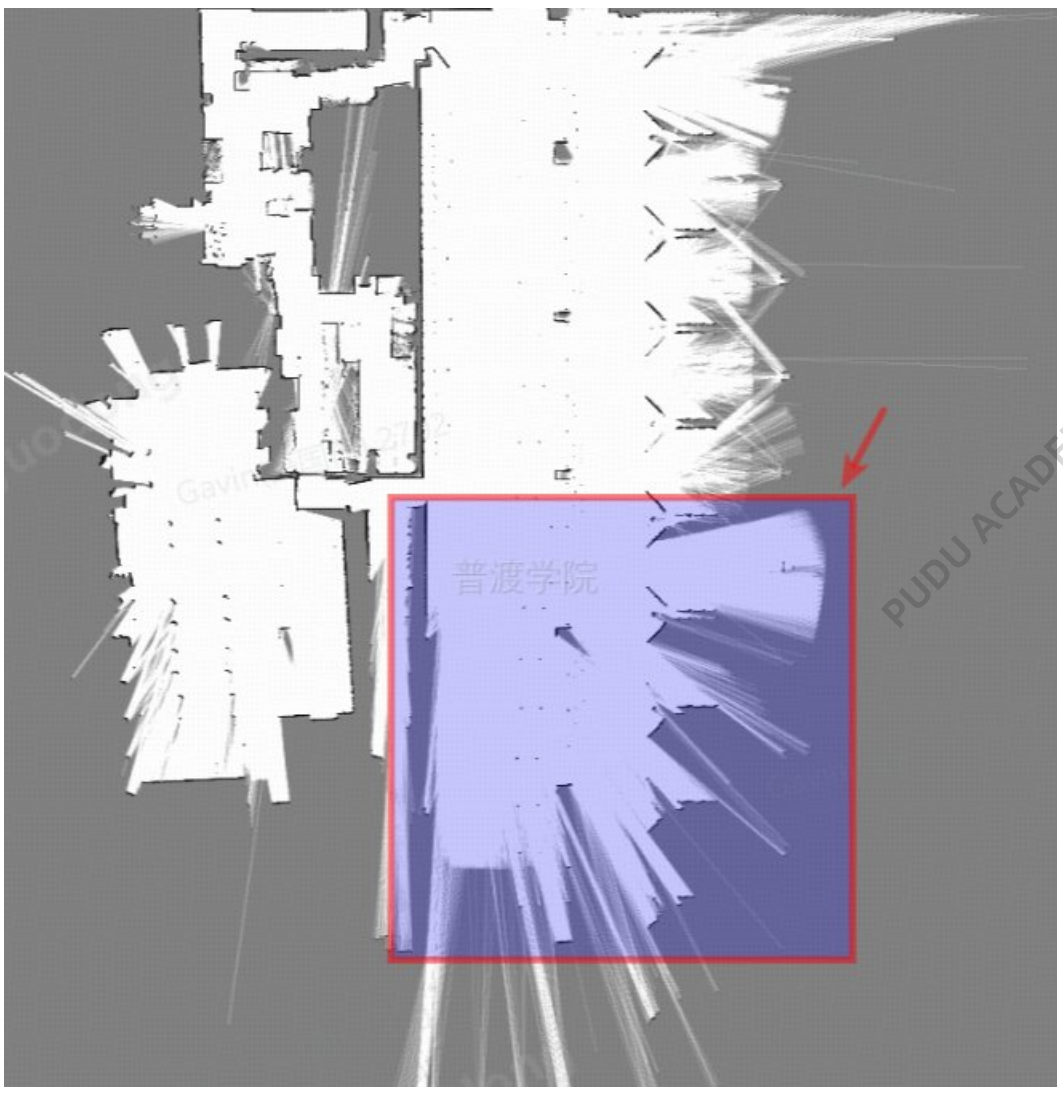
Illustration	Scenario	Mapping guidelines
	Restaurants and resorts: large maps, areas with fall risks, semi-outdoor environments, non-fixed decorations, hollow-back chairs, etc.	① First, map small areas, and then expand the mapping area using the "Extended Mapping" method; ② Startup points must be added at the boundary lines of each extended map; ③ In semi-outdoor areas, deploy virtual walls on both sides of the robot's travel path; ④ For maps with fall risks, add virtual walls and keep-out zone elements.

	Extra-large map: venue area of 5,000 m² to 10,000 m² or more.	① It is recommended to keep the single mapping area within 2000-3000 m². First, complete mapping of local areas, and then gradually expand the mapping area using "Extend mapping"; ② An activation point needs to be added at the boundary of each extended map. Note: segmented mapping can effectively reduce the data processing pressure of a single mapping, avoiding robot lag or abnormal positioning due to an excessively large map. If the total area of the venue exceeds 10,000 m², it is recommended to complete the extended mapping in 3–4 steps.

4.2.4 Common mapping problems and solutions

After mapping is complete, you need to check whether the map quality meets the standards. The table below lists common low-quality map problems. If any of the following situations occur, please handle them according to the corresponding solutions.

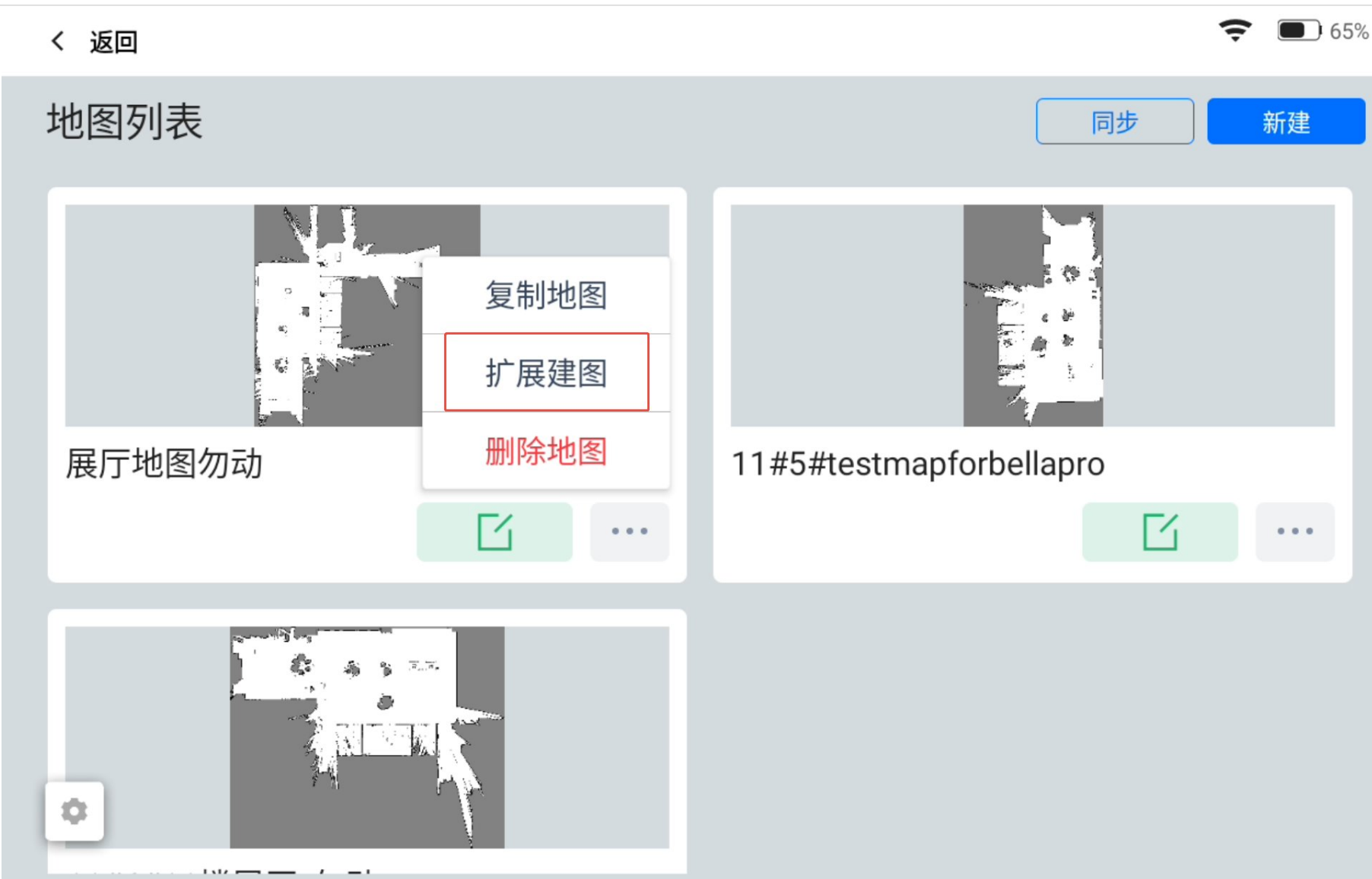
Example	Problem description	Risk	Solution
	Wall ghosting	Location hopping	Re-expand the ghosting area of the map.

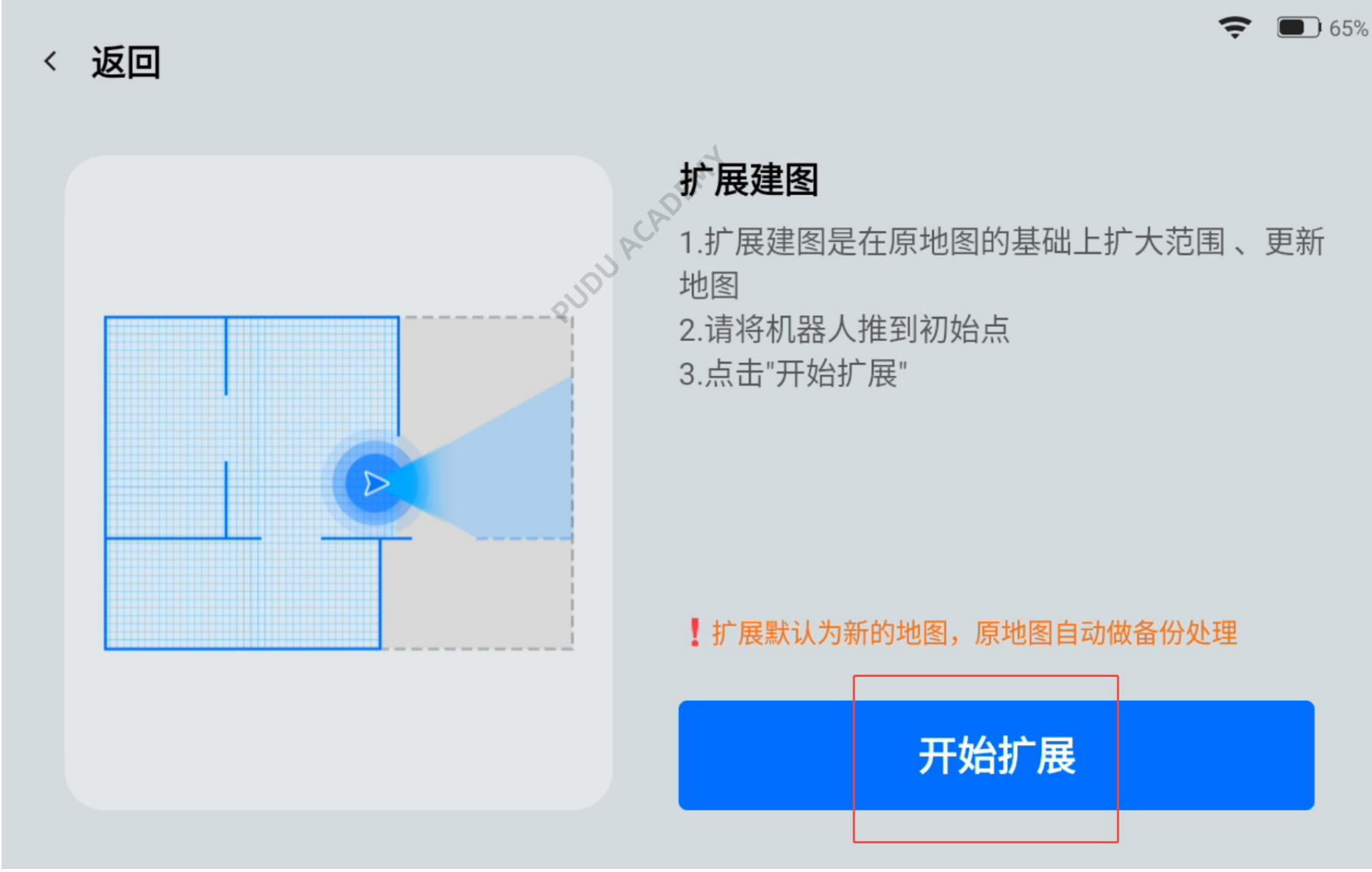
	Map distortion inconsistent with the actual environment	Loss of location	Re-map, and avoid passing through sloped areas during the pushing process.
	Incomplete map	Loss of location	Expand the map and ensure the edges are clear.

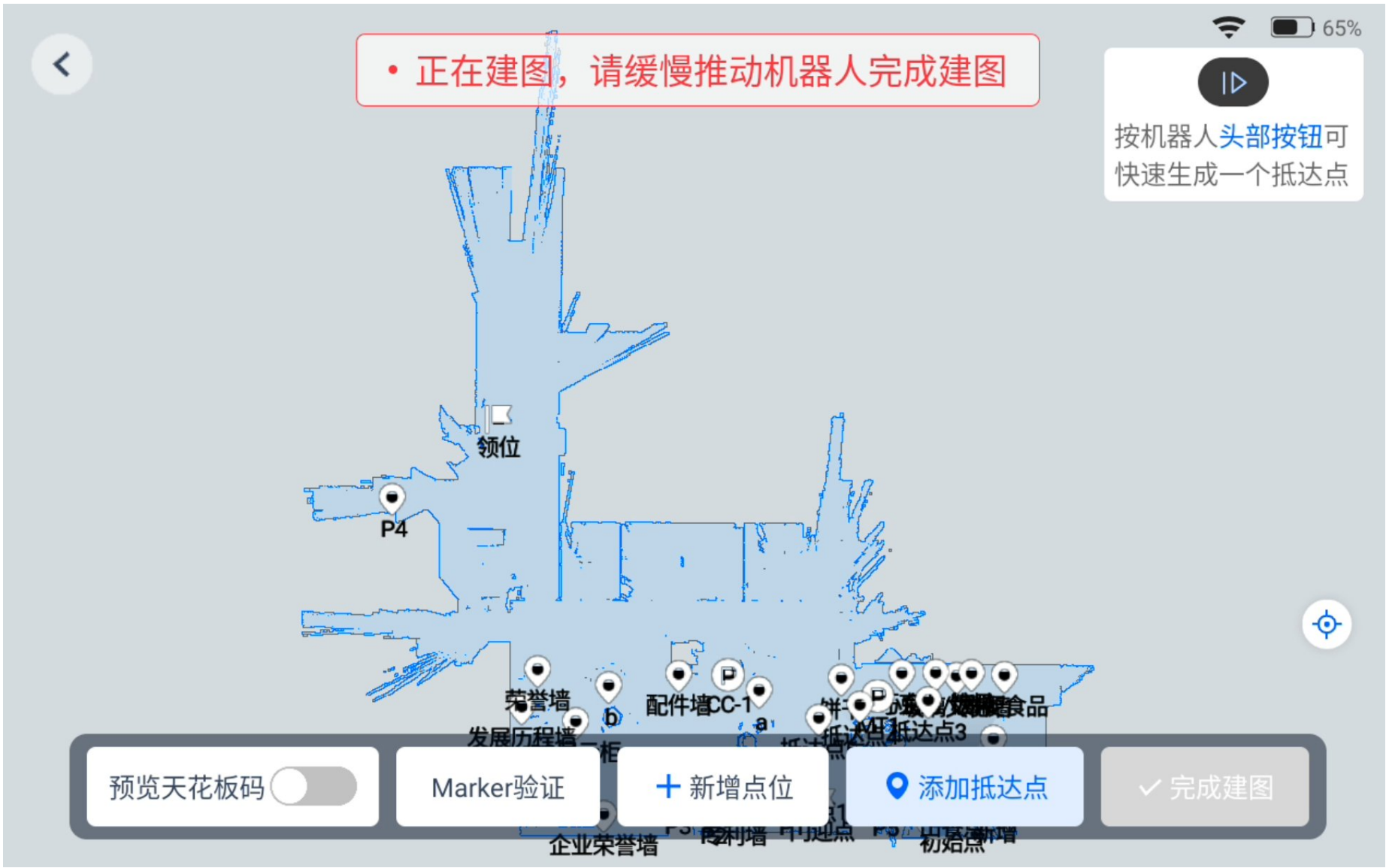
4.3 Extended mapping operations

When there are significant changes in the original map area or when map expansion is required, perform the "Extend mapping" operation. The extended mapping steps are as follows:

Push the robot to the power-on point. On the robot home page, select "Settings" > "Map settings" > "Edit map" to enter the map list interface. Click "..." > "Extended mapping" in the lower right corner of the target map to enter the map editing page. Click "Start extension."





Expand the map according to actual conditions, and save and upload after settings are complete.	

5.Edit map

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5.1 Erase noise points

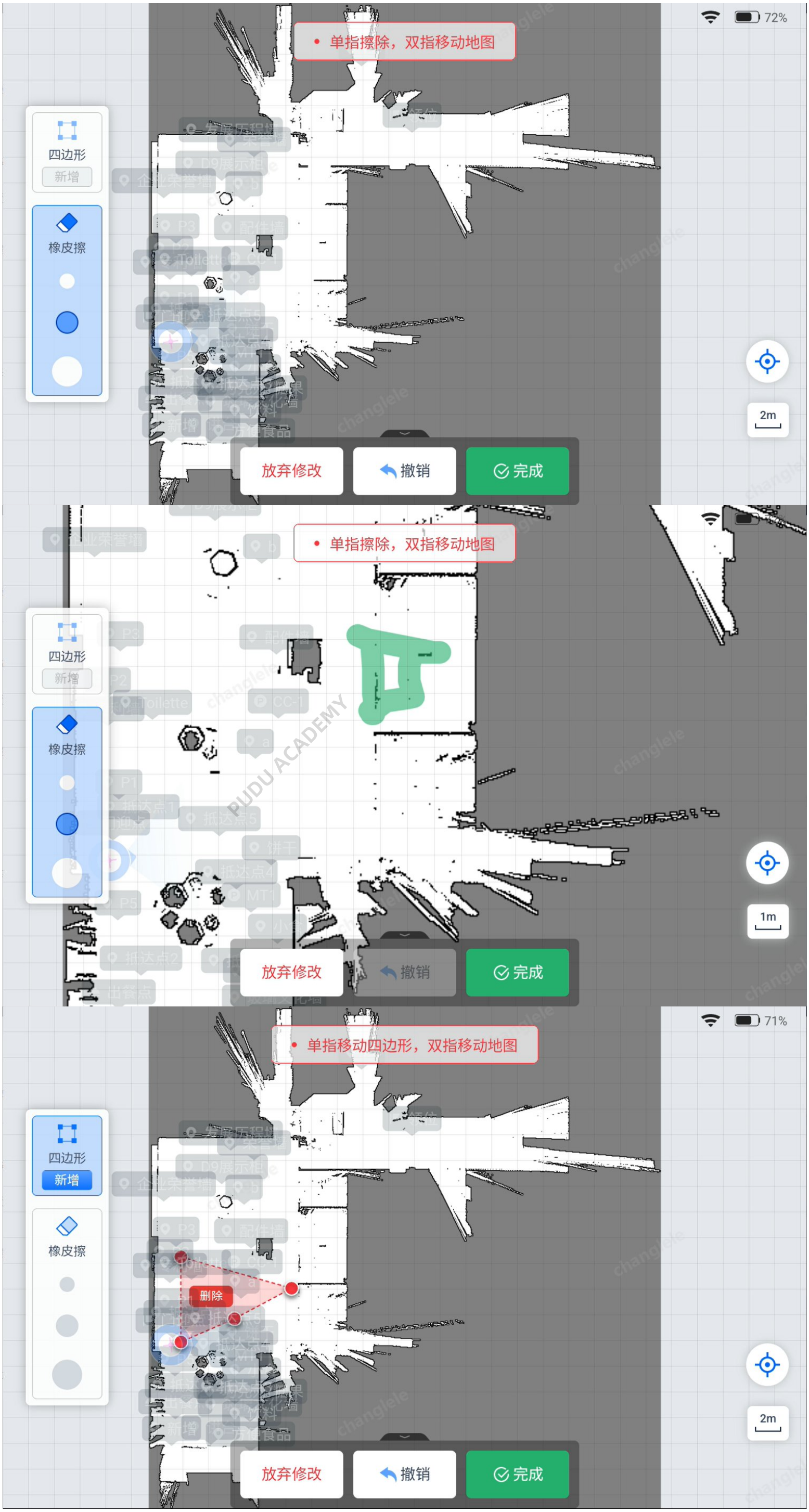
During the mapping process, if the robot passes a path with temporarily placed large obstacles (such as cardboard boxes, trash cans, etc.), the radar may identify them as fixed obstacles and record them on the map, forming noise points.
In this case, you need to use the manual noise point erasure function to clear the mistakenly recorded noise points. The operating steps are as follows:

Go to the Mapping tool > "Erase noise"	

There are two ways to erase: ① Eraser erase: used for small-area noise point erasure. ② Frame area erase: used for large-area noise point erasure.	

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Drag the map to the location where you want to erase noise points. Use the eraser tool or selection box to select the noise point area and complete the erasing operation. Click the undo button to re-select the noise point location to be erased.



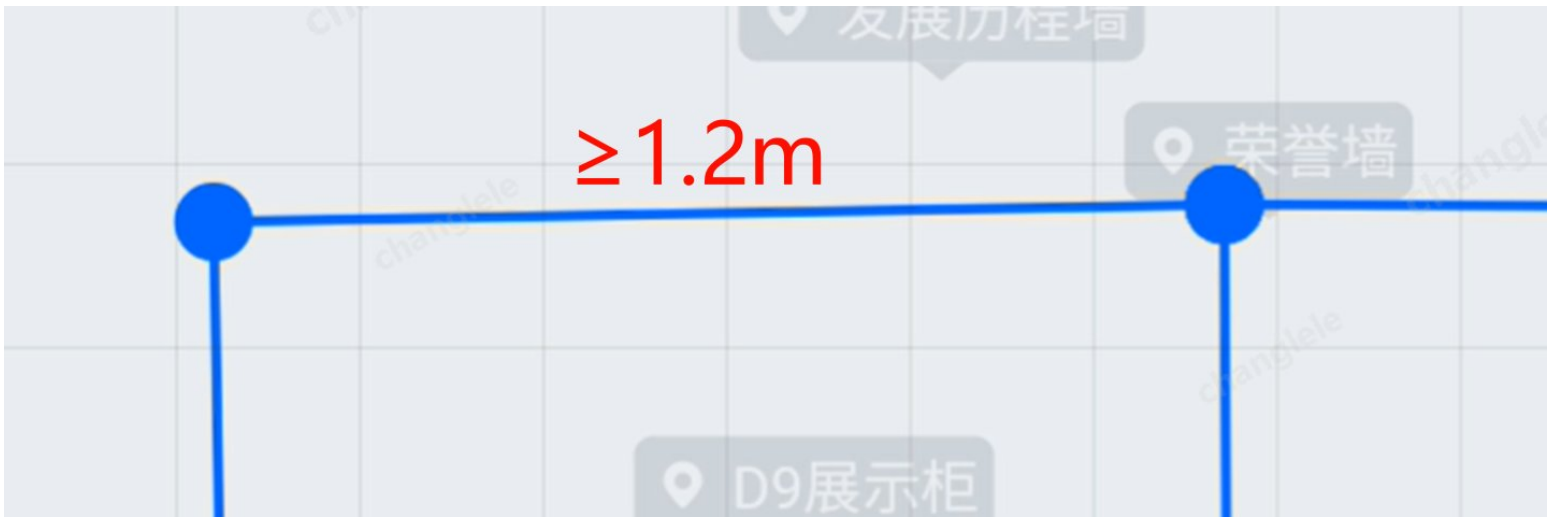
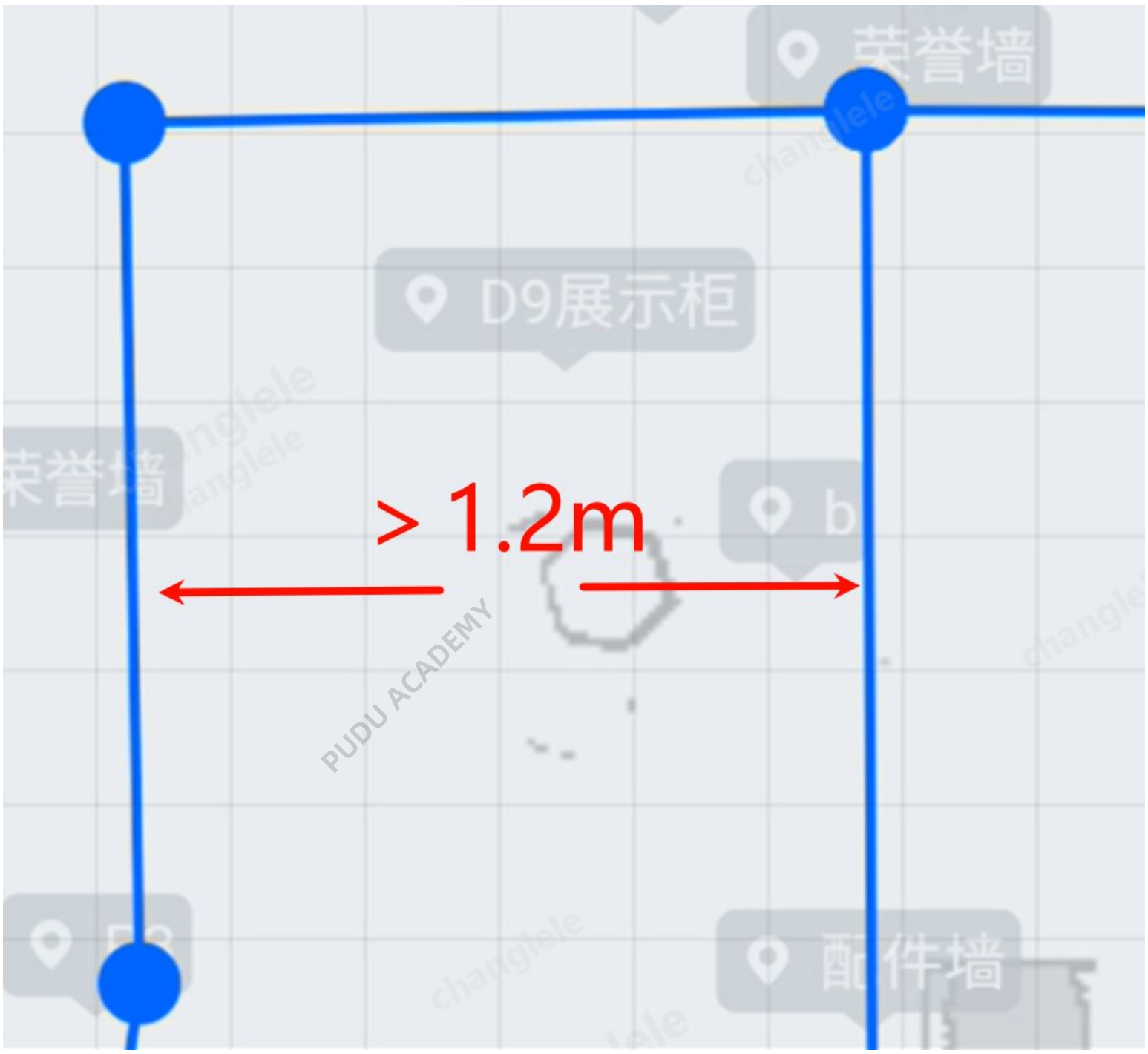
5.2 Topology path setup

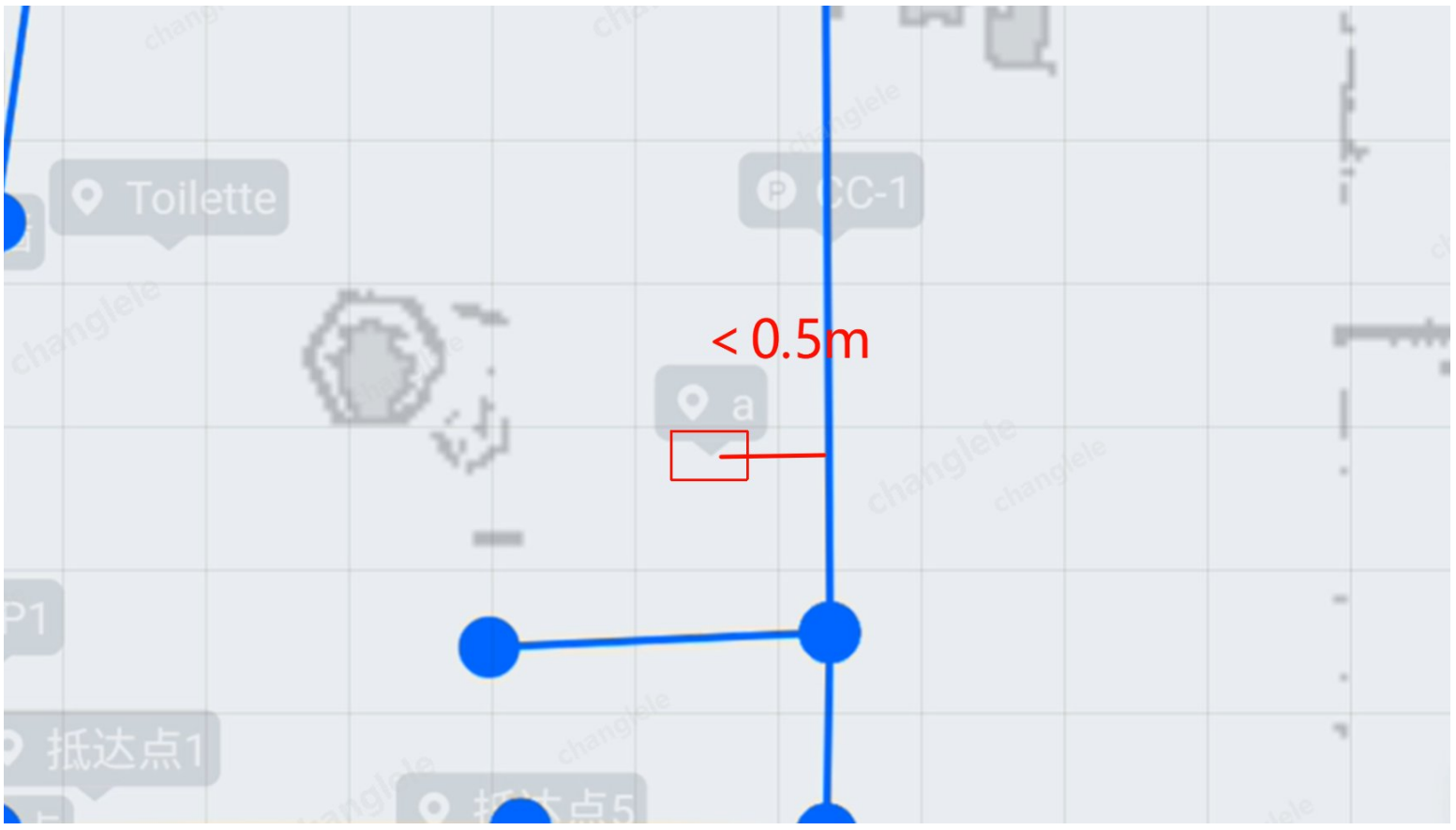
Topology paths can be considered the robot's planned driving paths. If no topology path is set in the target area, the robot will not be able to plan its entry into that area. You need to complete the topology path configuration to make the robot travel according to the preset route.

5.2.1 Precautions for drawing topology paths

Classification	Specific requirements	Remark

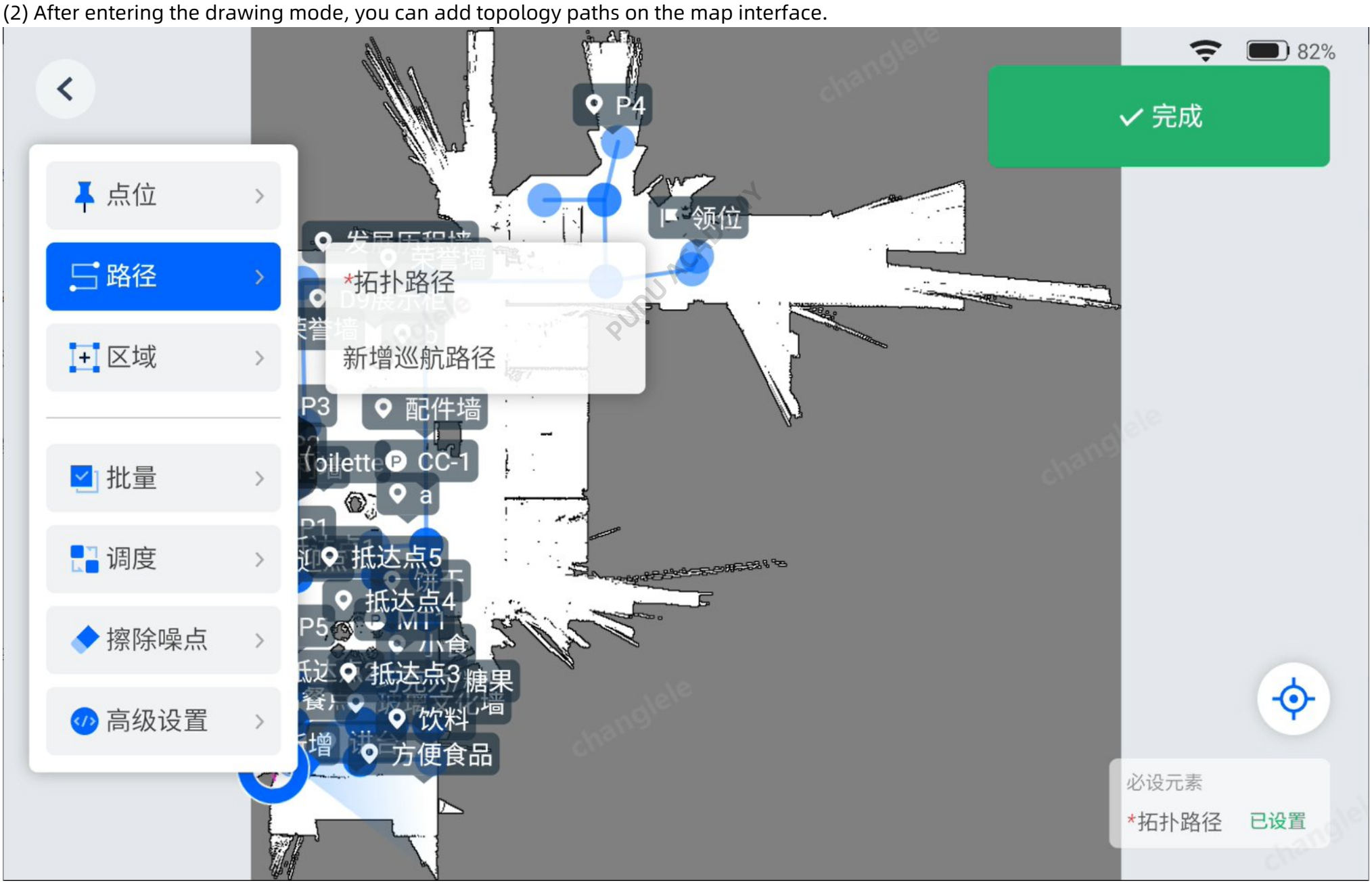
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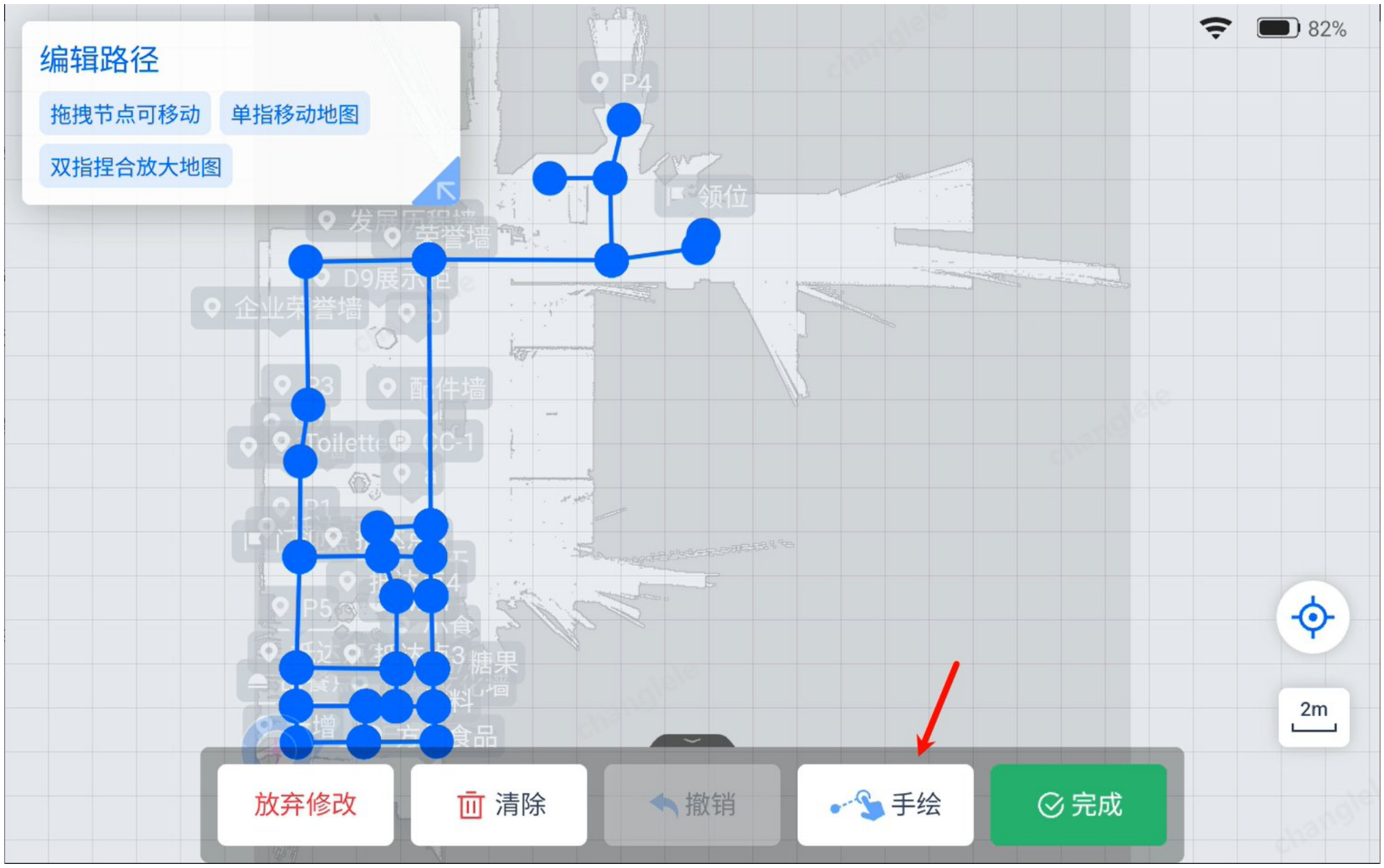
Path dimensions	Path length between two nodes ≥1.2 m	
	Adjacent path spacing >1.2 m	
	Path angle > 45°	

Point settings	Distance from point to path <0.5 m	

5.2.2 Draw topology paths

(1) On the operation interface, select "Path" > "Topology Path" > "Draw manually" mode.
(2) After entering the drawing mode, you can add topology paths on the map interface.

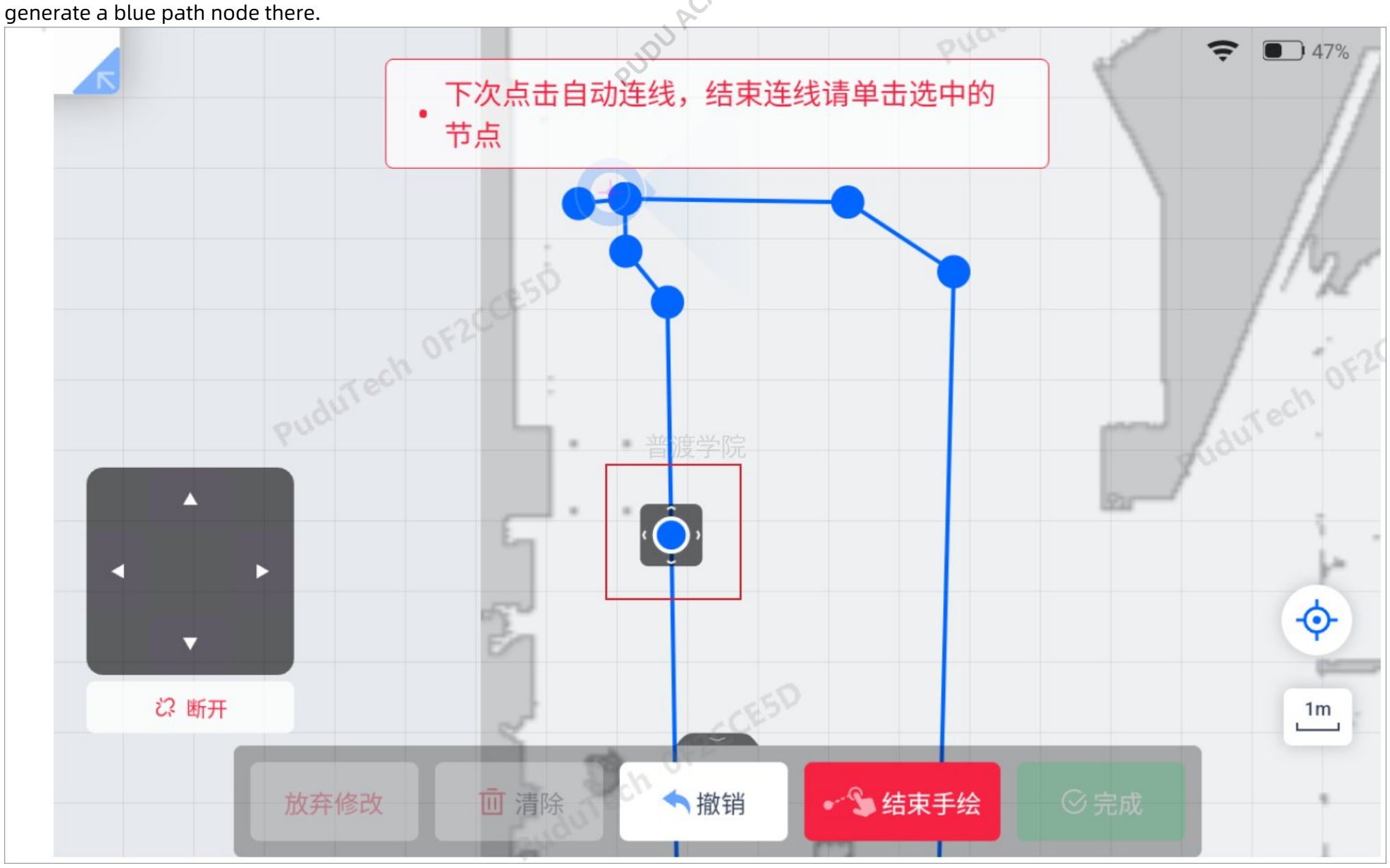




Note:

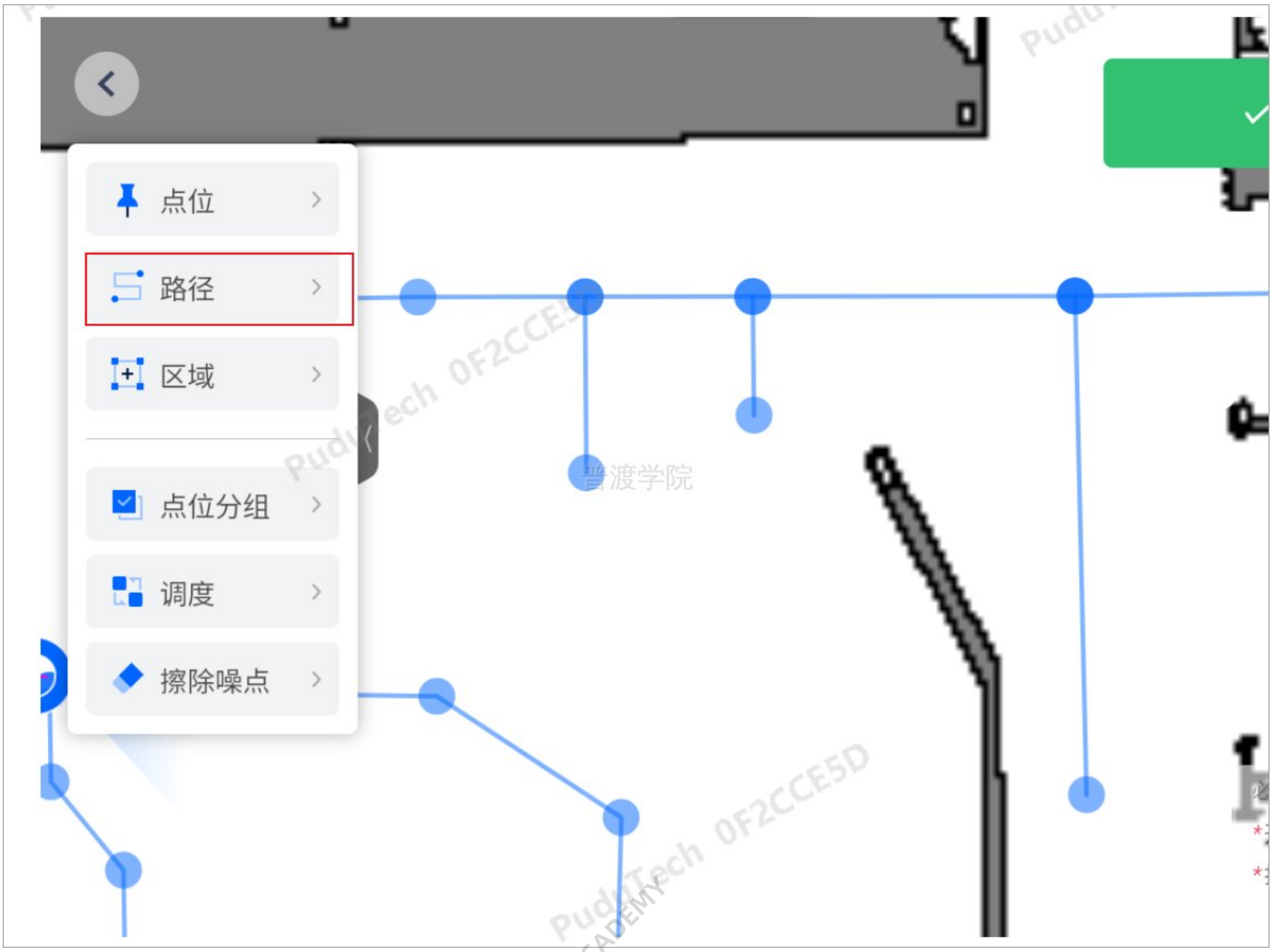
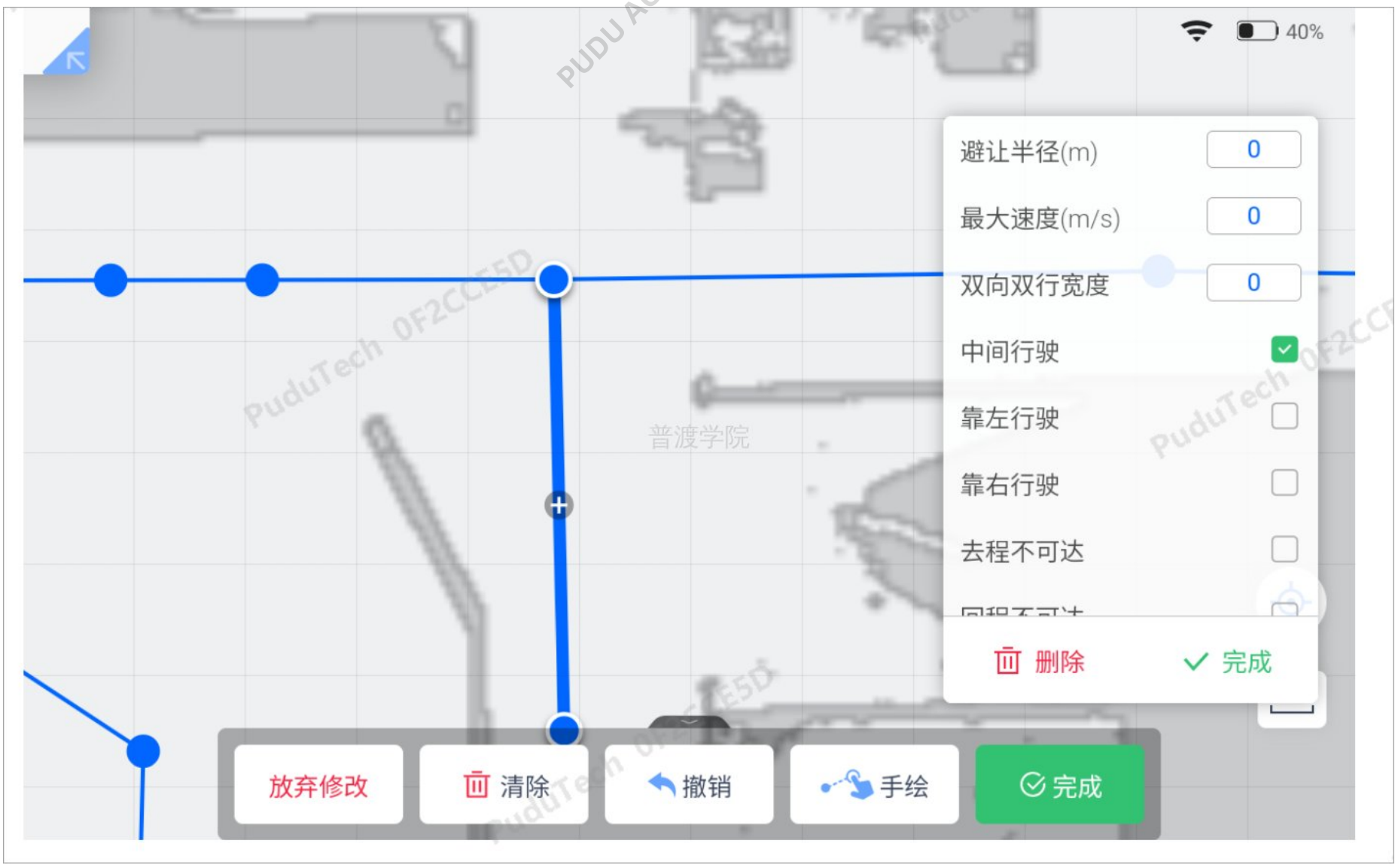
● If the robot loses localization, you need to manually push it to the startup point or mapping starting point to complete localization.

● In "Path" > "Topology Path" > "Draw manually" mode, directly tap on the map where you want to set a node. The system will automatically generate a blue path node there.



5.2.3 Topology path settings

How to set path properties:

Steps	Reference diagram
1. Select "Path" > "Topology Path".	
2. Select the path you want to constrain. After selection, the path will be displayed in bold.	

Path property options description:

Path options	Description
Avoidance radius (m)	The maximum distance the robot can offset when encountering an obstacle
Max speed (m/s)	The maximum speed allowed on this road segment

Bidirectional dual-lane width (m)	When the path width is >1.4 m, multiple robots are allowed to pass simultaneously. The value is the width of a two-way path, and the set value should be slightly less than the actual width.
Center driving	The robot will drive in the middle as much as possible.
Keep left	The robot will keep to the left as much as possible.
Keep right	The robot will keep to the right as much as possible.
Inaccessible outbound	The robot does not pass through this path when reaching its destination.
Return trip unreachable	The robot does not use this path when returning to the destination.

Bidirectional dual-lane width setting

1. Bidirectional dual-lane width overview

If multiple robots need to perform tasks on the same road section, set up bidirectional dual-lanes to allow two robots to pass simultaneously on a single road section, improving passing efficiency and system intelligence.

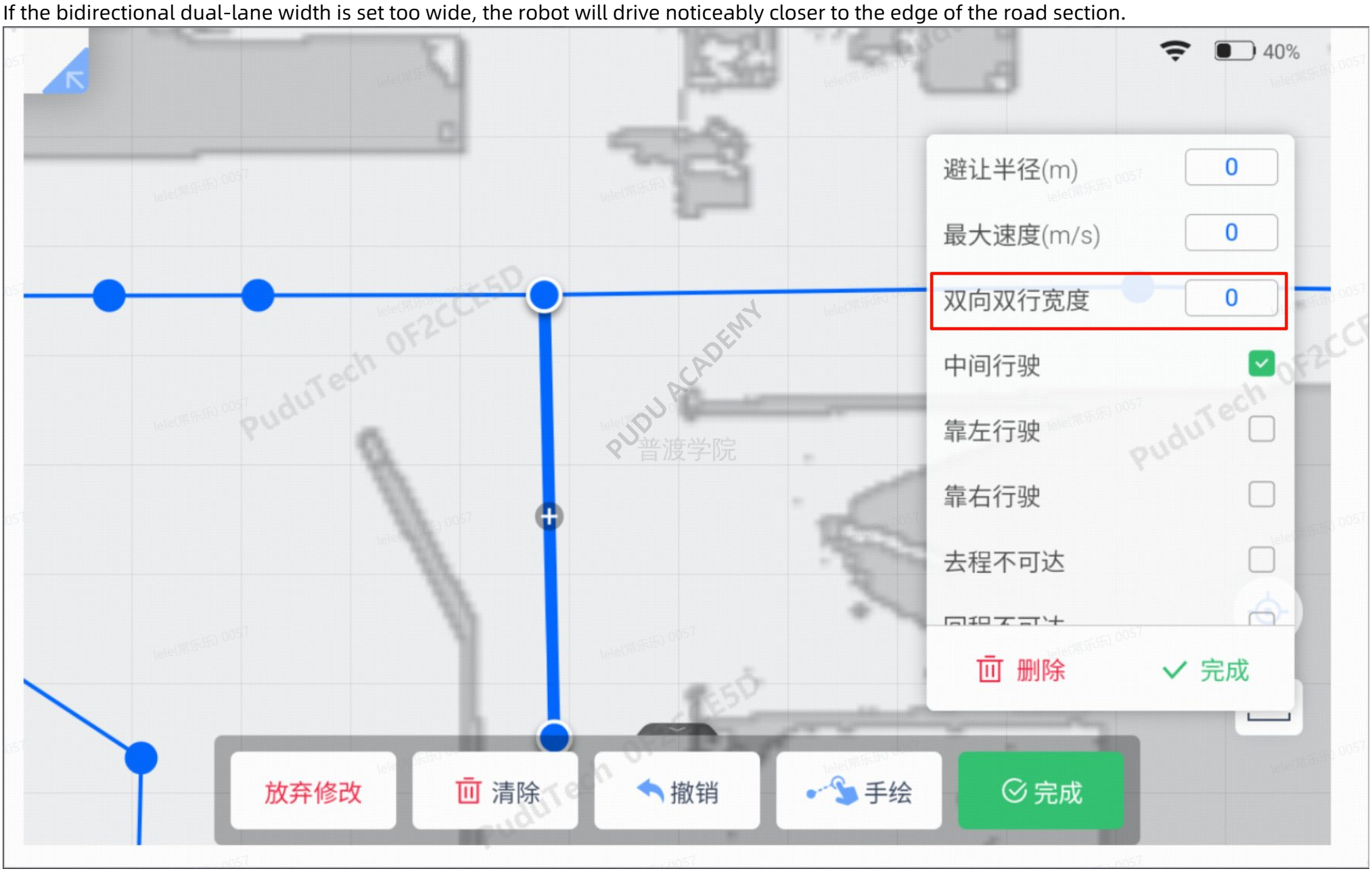
2. Set bidirectional dual-lane width

1. Confirm that the actual width of the road section allows two robots to pass at the same time (the minimum bidirectional two-lane width required for robots is 1.4 m).

② Select the path to be set, and configure the bidirectional dual-lane width according to the actual site conditions.

Must meet the following: 1.4 m < two-way two-lane width < actual road section width.

If the bidirectional dual-lane width is set too wide, the robot will drive noticeably closer to the edge of the road section.



③ After entering the bidirectional dual-lane width, click "Done" to complete.

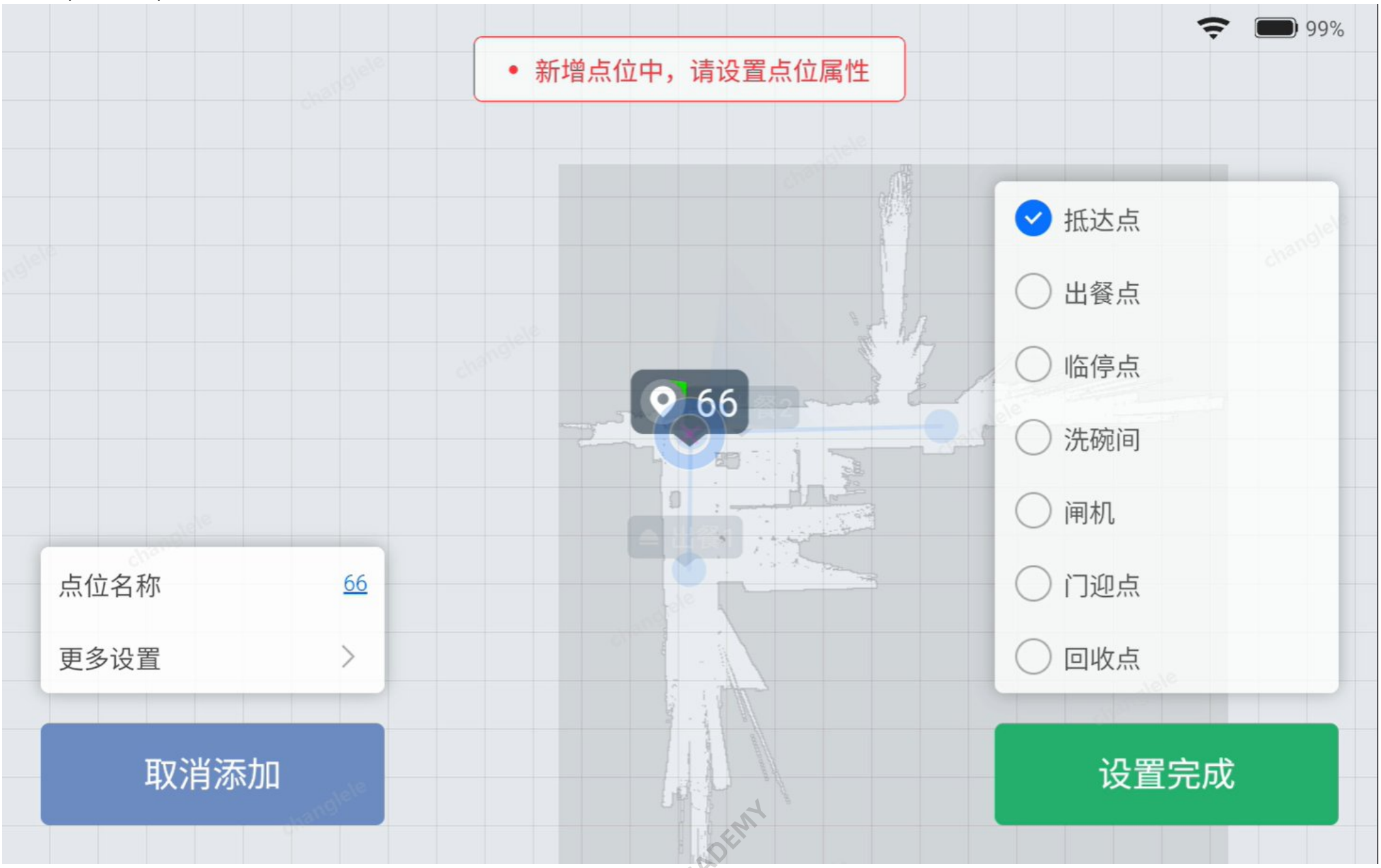
5.3 Add points

5.3.1 Common points

Point type	Property description
Arrival point	Robot delivery items arrive at the target location
Serving point	The food-serving point is a fixed location where cooked dishes are sent out from the kitchen, centrally placed, and await pickup; it is also the location where the robot returns and docks after completing a task.
Power-on point	After the robot is powered on or loses its localization, the operator pushes the robot to this position to complete localization.

5.3.2 Add points

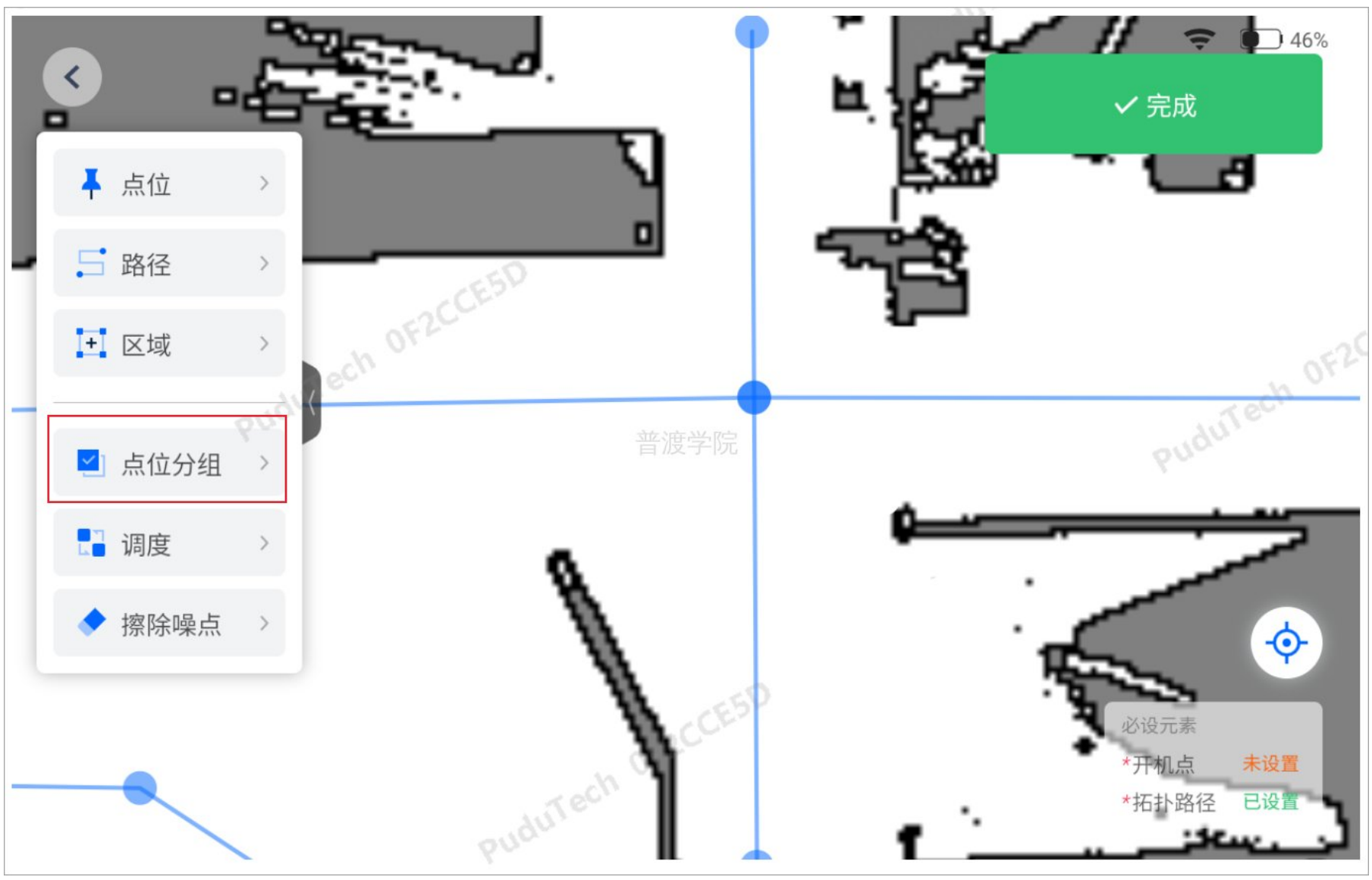
After successful localization, select "Point" and push the robot to the target position. After selecting the corresponding point type, select "Done" to complete the point creation.



5.3.3 Add point groups

If the customer needs to manage a large number of points and has difficulty issuing tasks, they can use the point grouping function. This allows for bulk task issuance by group, simplifying the operation process.
Both arrival points and pick-up points support point grouping, and the grouping operations for both types of points are identical. Here, we use arrival points as an example to explain the specific method of point grouping:

① Select "Point grouping";



② Select "Add group" and set the group name;



③ Drag single/multiple arrival points into the group or select "Add arrival point" directly to finish adding points;



④ After grouping, you can directly select the group and then the corresponding task for the arrival point in the task interface.

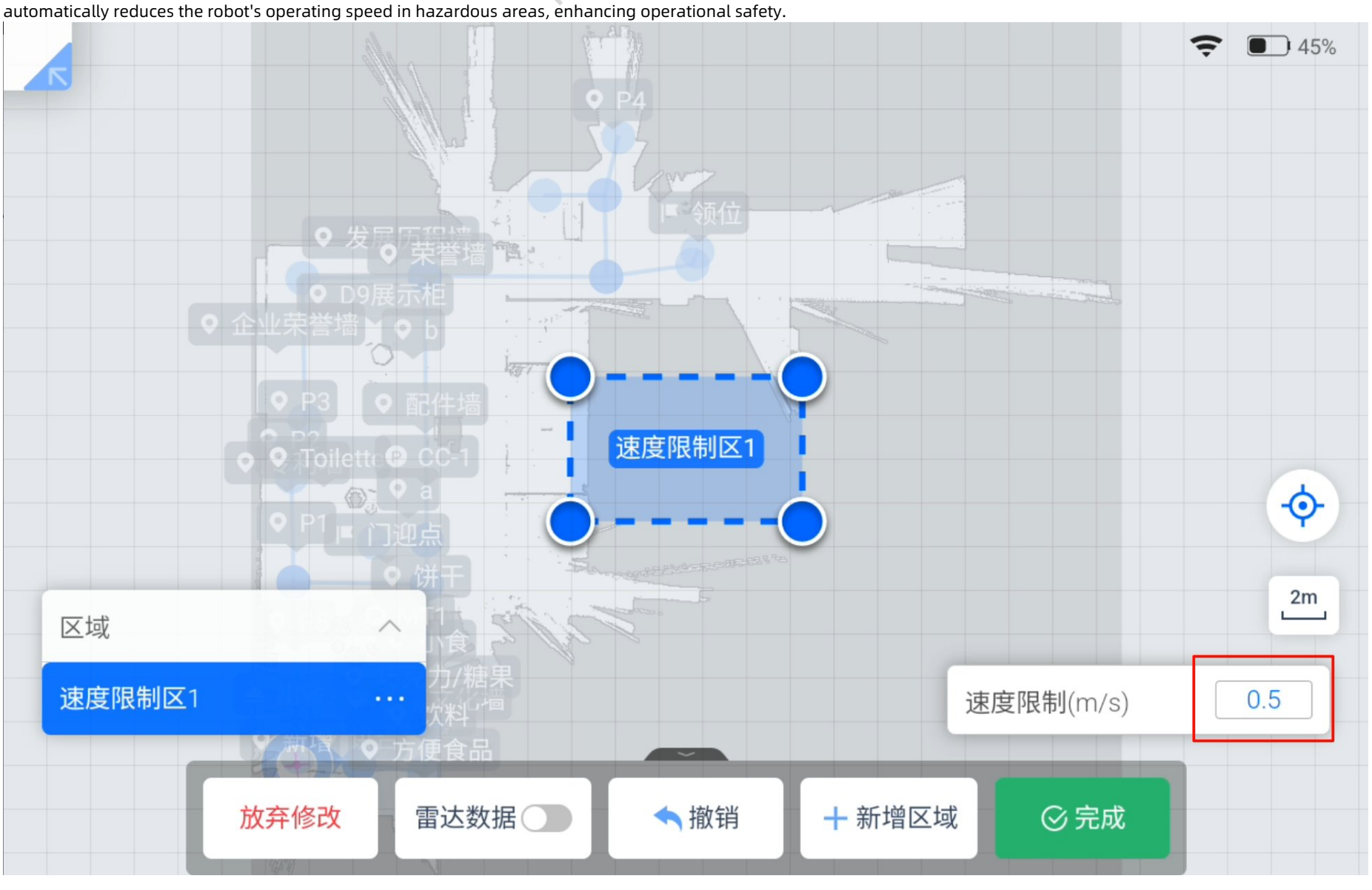


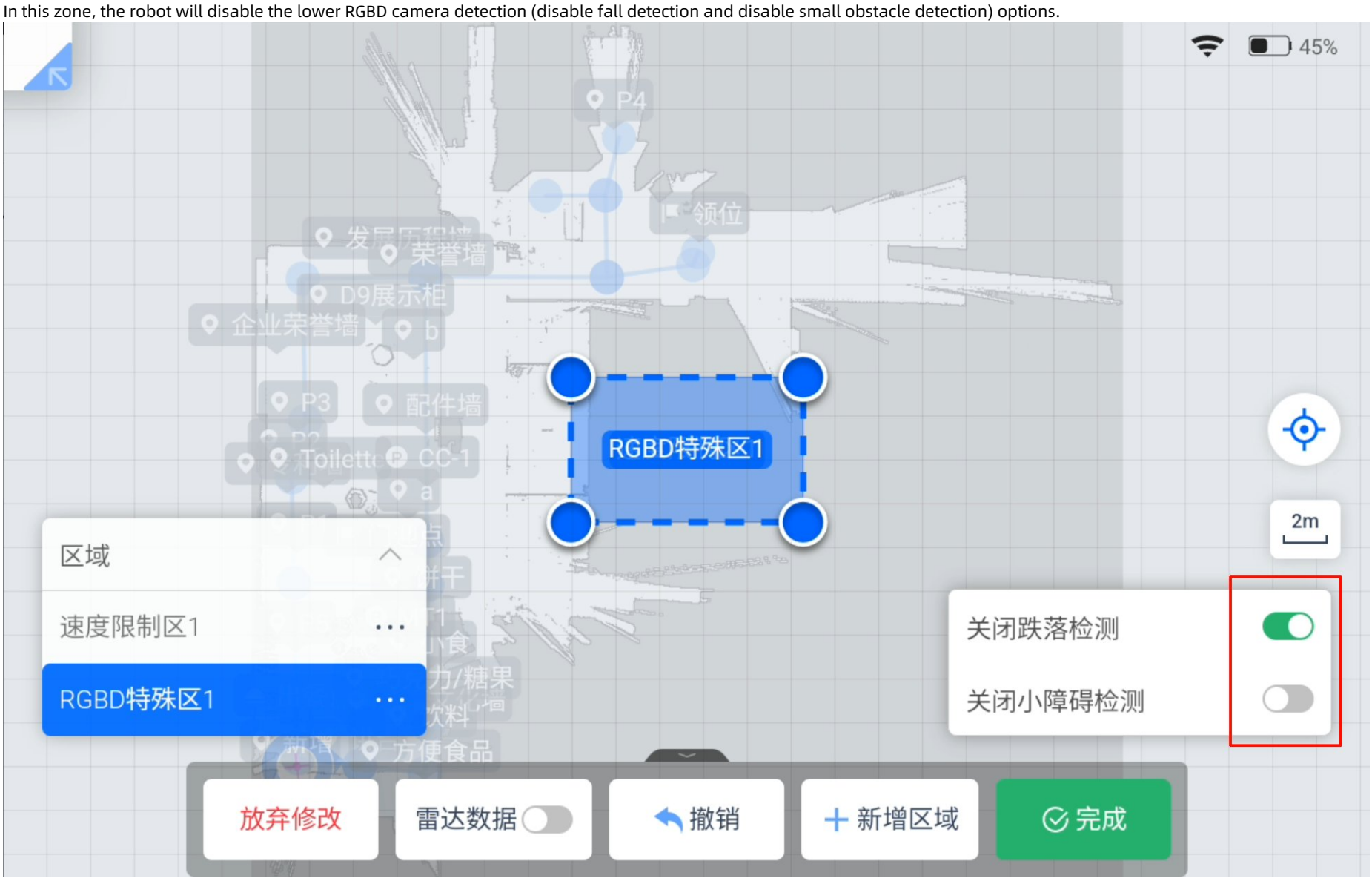
5.4 Zone settings

5.4.1 Common zones

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No-detour zone	After the robot reaches this position, it will travel along a preset fixed path without performing obstacle avoidance, turning, or detouring maneuvers. This strategy ensures operational safety, prevents collisions with equipment or blocked passages, and guarantees stable delivery task completion.
Speed limit zone	You can restrict the robot's operating speed (0.2–0.9 m/s) within designated areas. This limit takes effect when the robot's set speed is lower than the task speed. This measure automatically reduces the robot's operating speed in hazardous areas, enhancing operational safety.



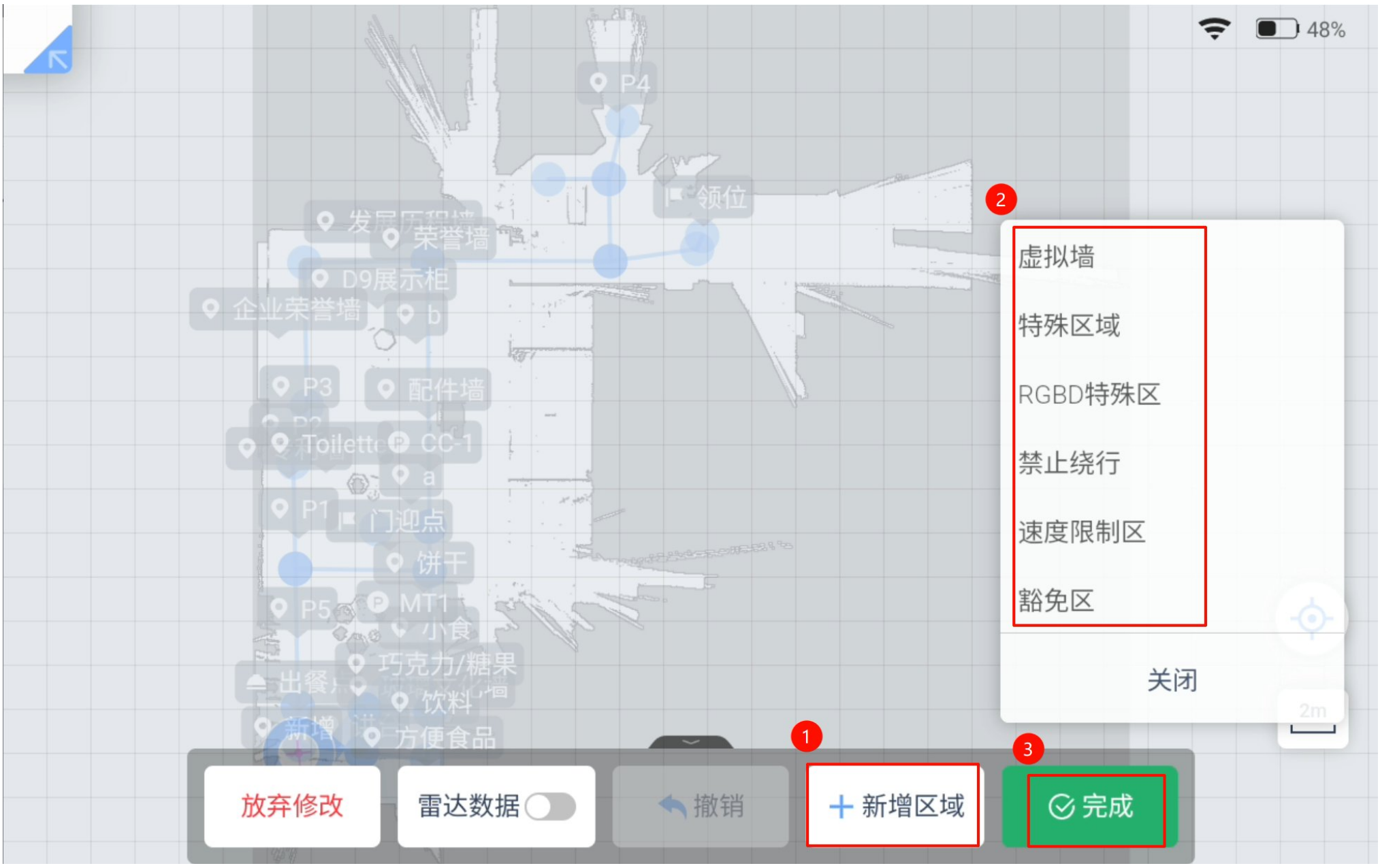
RGBD special zone	In this zone, the robot will disable the lower RGBD camera detection (disable fall detection and disable small obstacle detection) options. 
Virtual wall	When a robot's path contains areas with safety risks or restricted areas, you can set up virtual walls to make the robot avoid these paths.

5.4.2 Add zone

(1) Areas > Edit area

(2) Select the area to add, then select "Complete."

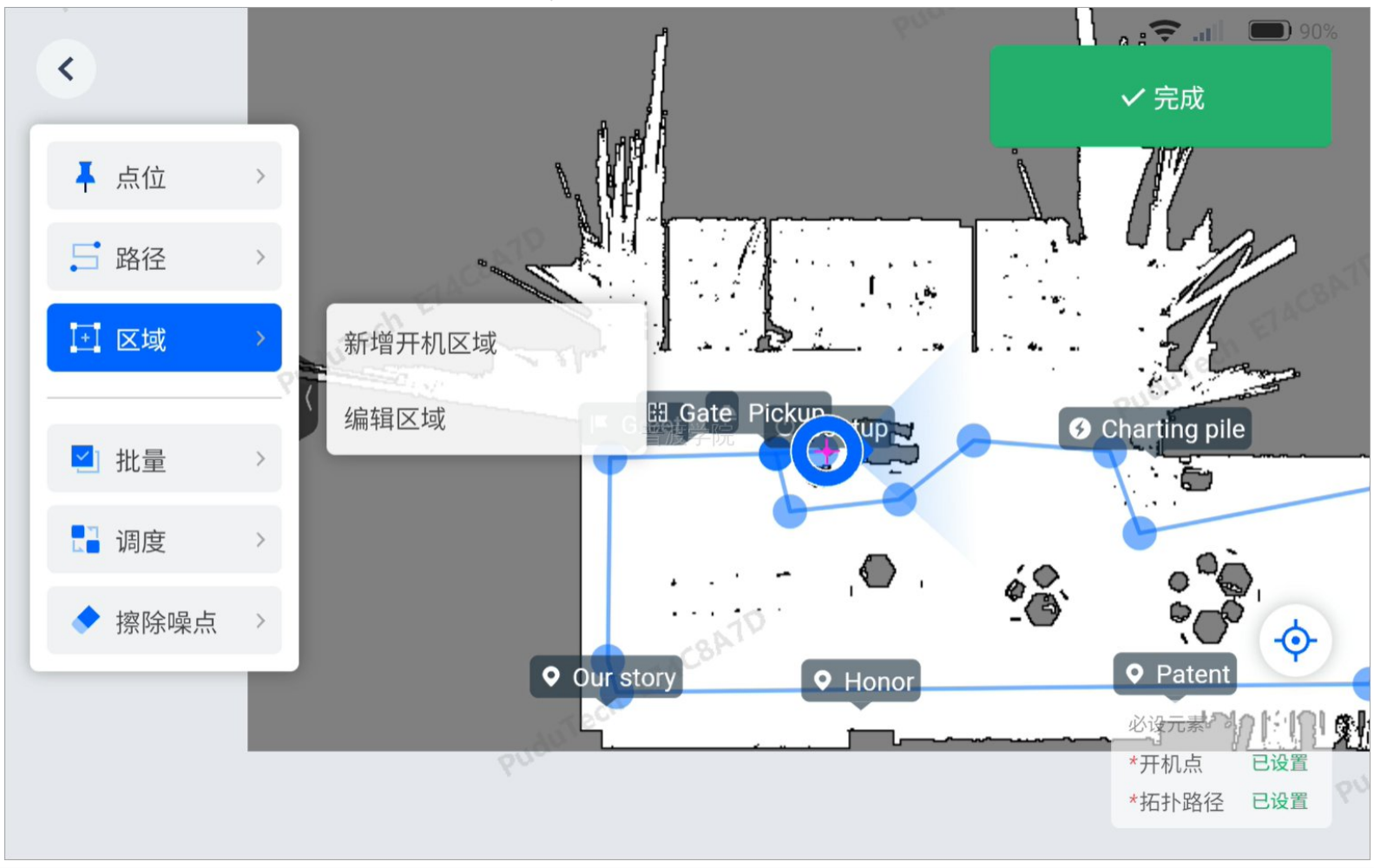


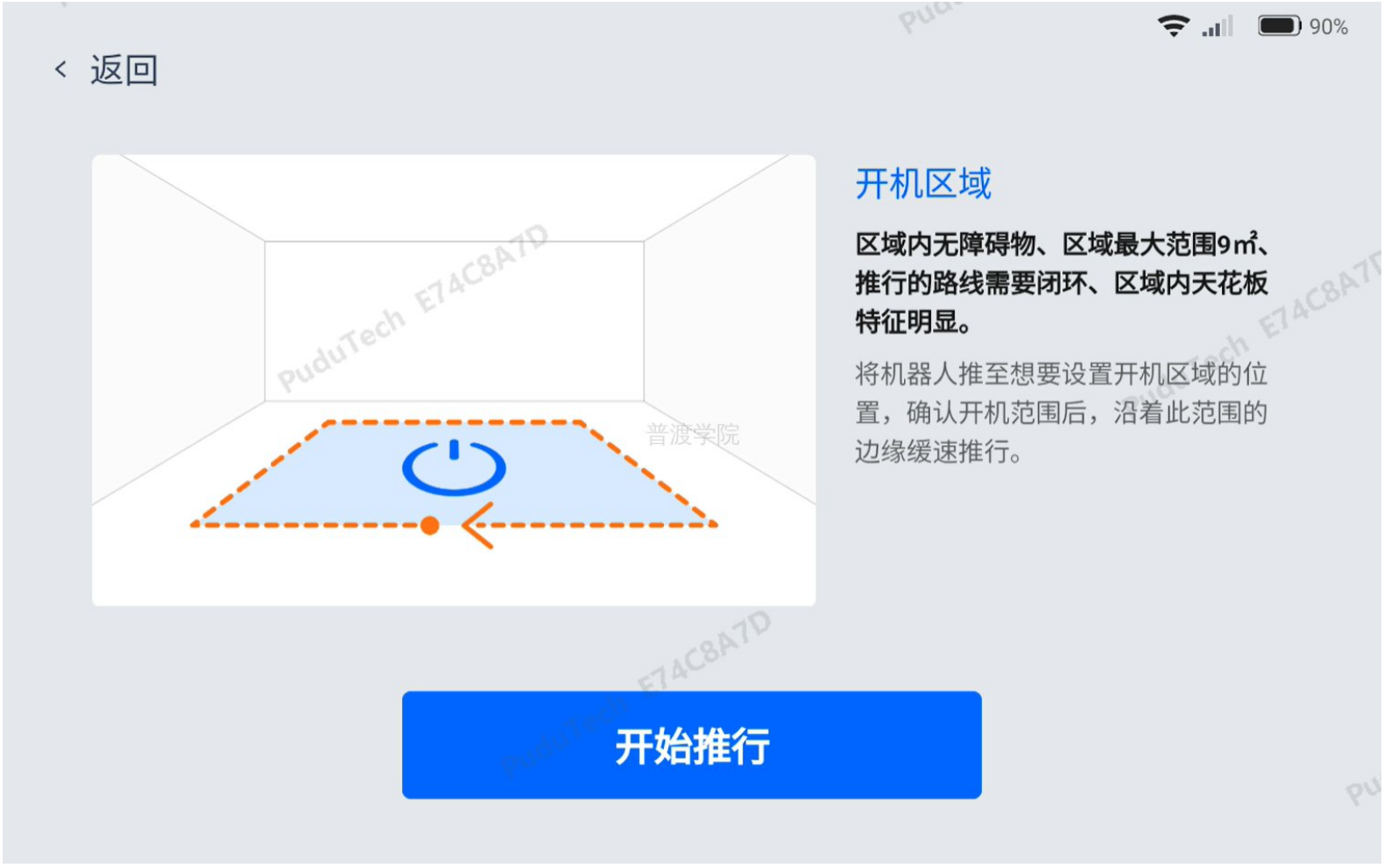
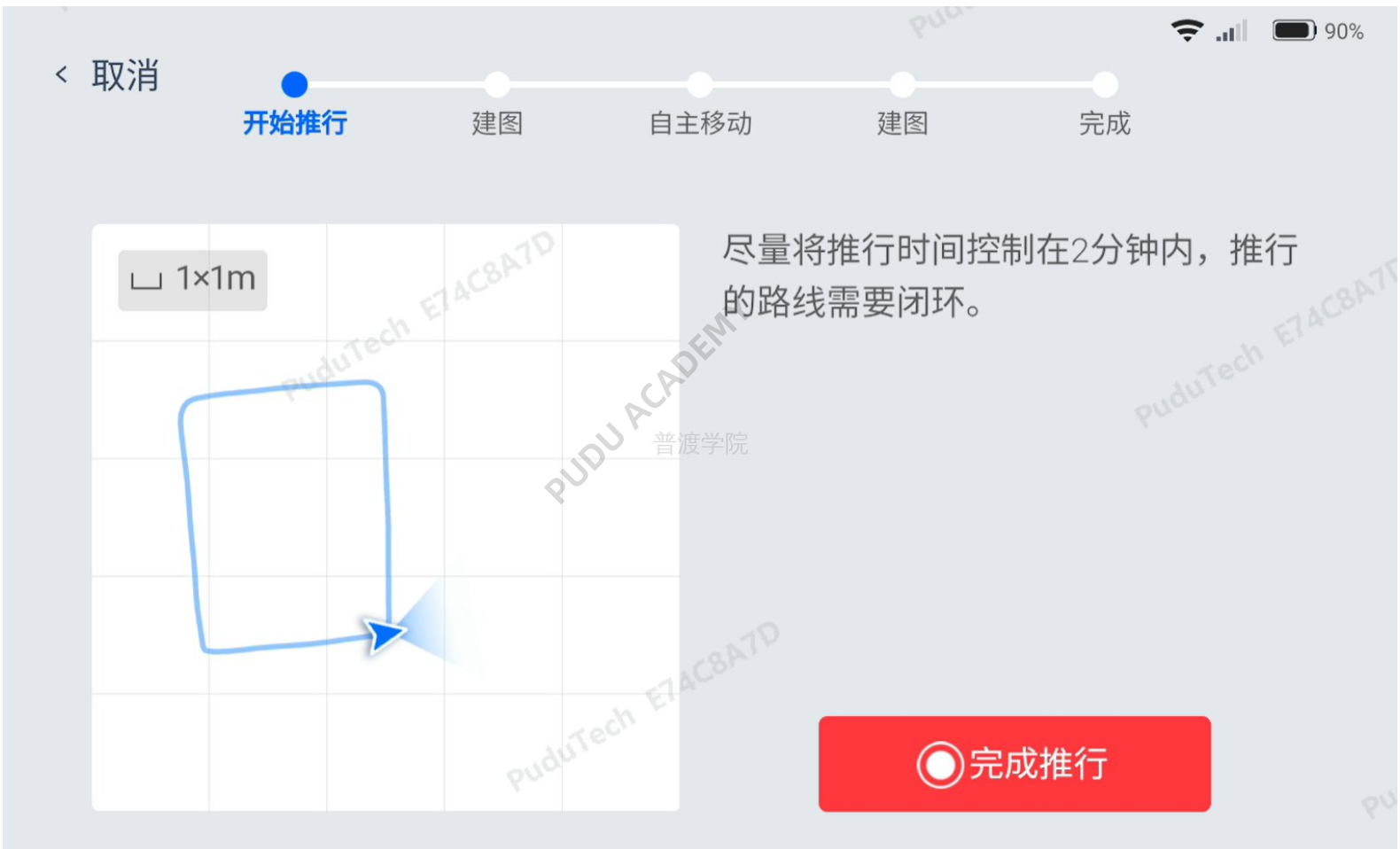


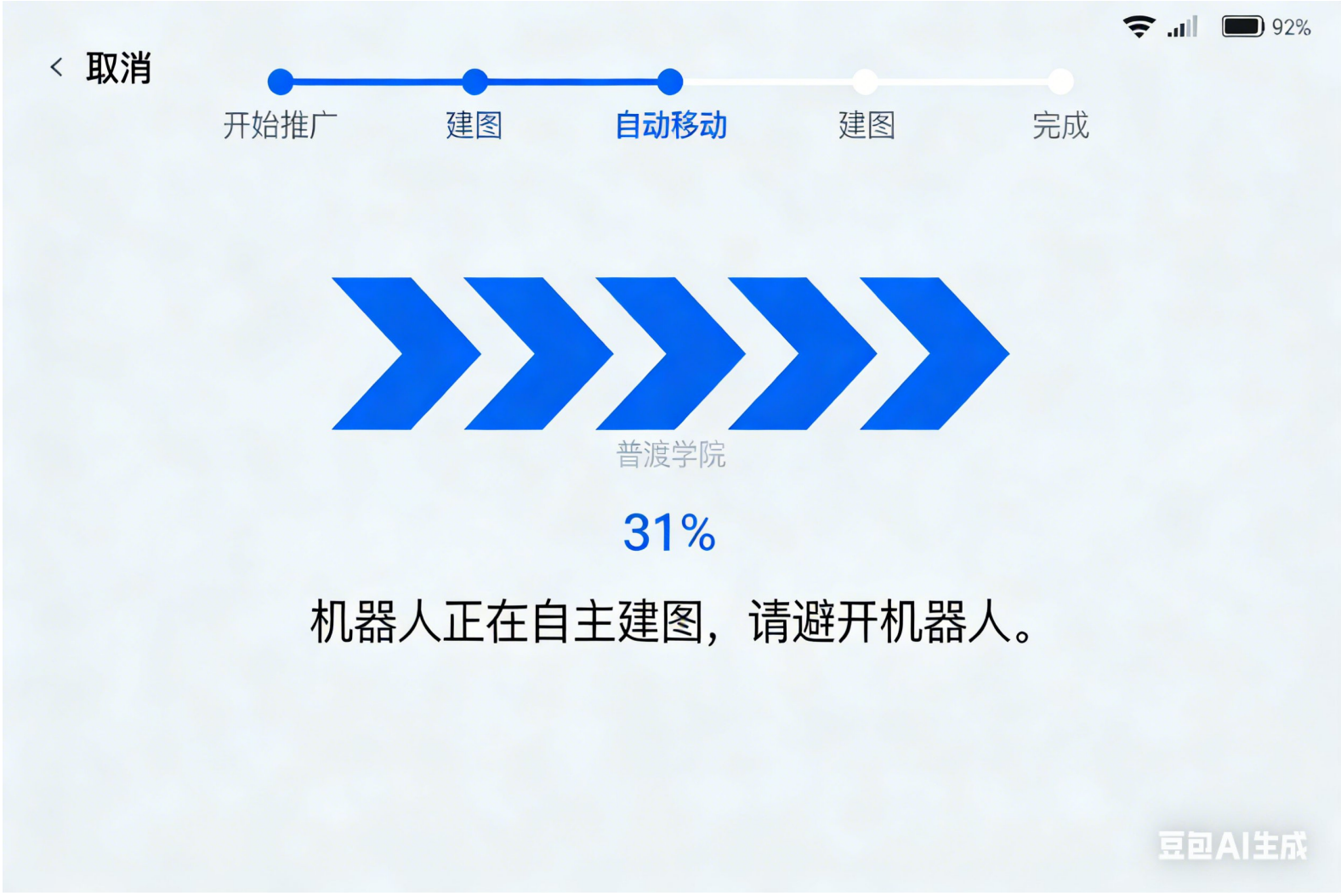
5.4.3 Bootup zone

The startup region is more conducive to the robot's successful positioning than the startup point. Simply push the robot into the startup region, and positioning will be completed without needing to push it to the startup point. Only one startup region can be added per map.
The steps to add a bootup zone are as follows:

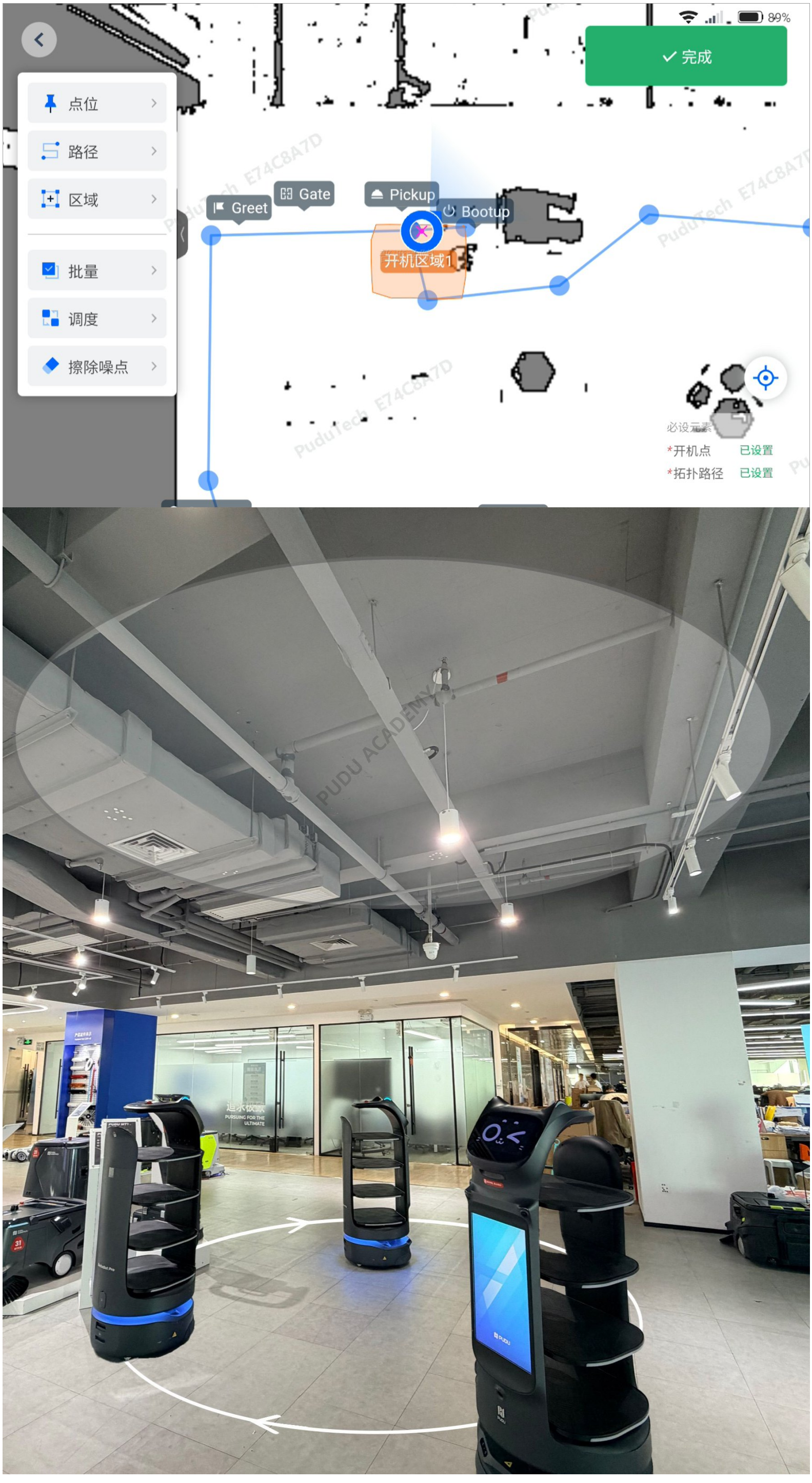
1. Select "Region" > "Add startup region."



2. Push the robot to the target startup region, and push it slowly along the edge of this region. Startup region requirements are as follows: ● The startup region must be free of obstacles ● The maximum area of the region is 9sqm ● The ceiling features in the region must be clear	
3. After manually pushing the robot to complete the path loop along the edge of the region, select "Finish pushing."	

4. Wait for the robot to finish autonomous movement. The robot will not flexibly avoid obstacles during autonomous movement. Please pay attention to on-site safety and ensure that there are no pedestrians or obstacles in the startup region.	
5. Wait for the robot to complete mapping. Please keep the robot stationary during local mapping.	

6. After the startup region is completed, the map will display the corresponding orange-yellow region box.



5.5 Safety settings

5.5.1 Risk zone deployment

Risk scenarios	Example	Deployment requirements
Ramps		① Set path speed to 0.3 m/s for uphill/downhill sections. ② Set uphill/downhill areas as exemption zones. ③ Set uphill/downhill areas as no-bypass zones. Testing requirement: The robot must continuously traverse uphill/downhill 10 times to ensure normal passage.

Stairwell	 	① Set up a virtual wall at the stairwell entrance that covers the entire entrance. ② Set up a no-bypass zone in the aisle next to the stairwell. Test requirement: Manually block the robot's direct path and conduct 10 consecutive tests. The robot should pass safely every time.

Glass wall surface		Glass environment in corridors: ① Virtual walls should be added to all glass areas. ② No-bypass zones should be added to areas next to glass walls. Test requirement: Manually block the robot's direct path and conduct 10 consecutive tests. The robot should pass safely every time.

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5.5.2 Fall prevention deployment

The robot offers three anti-fall detection methods: RGBD, reflective strips, and magnetic strips (optional). It is recommended to prioritize the RGBD anti-fall solution. Anti-fall deployment needs to be coordinated with "speed limit zones" to reduce the robot's running speed, thereby improving its recognition accuracy and reliability for fall areas, and ensuring the robot can promptly and accurately identify and take protective measures.

5.6 IoT settings

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5.6.1 Turnstiles

Before adding a turnstile, you must deploy the turnstile hardware module.
When deploying a gate point, the robot should pass through the center of the gate or directly below the automatic door, and the robot's orientation must be consistent with the gate point's direction of passage.



① On the map editing page, select "Points" to go to the point settings page;
② Select "Add point";
③ Select "Turnstile" as the point type, add a turnstile point at the robot's current position, and bind the corresponding device ID.



Select "More settings" on the page to adjust parameters according to the actual scene.
Upstream: The direction the robot enters the turnstile.
Downstream: The direction the robot exits the turnstile.



Uphill deceleration zone distance: This parameter defines the area where the robot enters the turnstile. After entering this distance range, the robot will decelerate and begin to establish communication with the turnstile.
Uphill deceleration zone speed: After the robot enters the "uphill deceleration zone distance," it will travel at this speed (if other speed limits exist, the minimum value will be taken).

	Uphill stop zone distance: When the robot's distance to the turnstile reaches this set value, it will send an open door command, lock the motors, and stop in front of the turnstile, waiting for a signal from the turnstile. Note: The uphill stop zone is a minimum of 1 m. If the set value is less than 1 m, it will be effective at 1 m.

	Downward stopping distance After the robot passes the gate's center point, the system will stop recognizing that gate. If the robot has not fully passed the gate, the system will determine whether the robot has fully exited based on a preset distance, and will disconnect communication with the gate after confirming full exit.

For information about the elements of the mapping tool, please refer to the Pudu Robotics Academy.

6. Robot testing

6.1 Business test

After map deployment, you need to observe the robot's operation through test runs and adjust and optimize the map based on actual operating conditions. The specific test requirements are as follows:
1. Test all points to check the rationality of path planning and the accuracy of point arrival.
2. For areas with significant scene changes and areas outside the map construction path, to ensure robot positioning accuracy, test runs need to be conducted to verify whether the robot can autonomously identify the latest map environment (VSLAM and laser will automatically complete map updates).

6.2 Peak-hour testing

Some areas may experience peak passenger flow during specific times. By conducting tests during peak hours, you can identify links where robot operating efficiency decreases. Based on on-site needs and environmental characteristics, the following adjustments can be made:
(1) Adjust the robot's running route to align with the movement paths of the on-site personnel
(2) Adjust to align with the customer's on-site business operations
(3) Adjust robot function configuration according to scene requirements